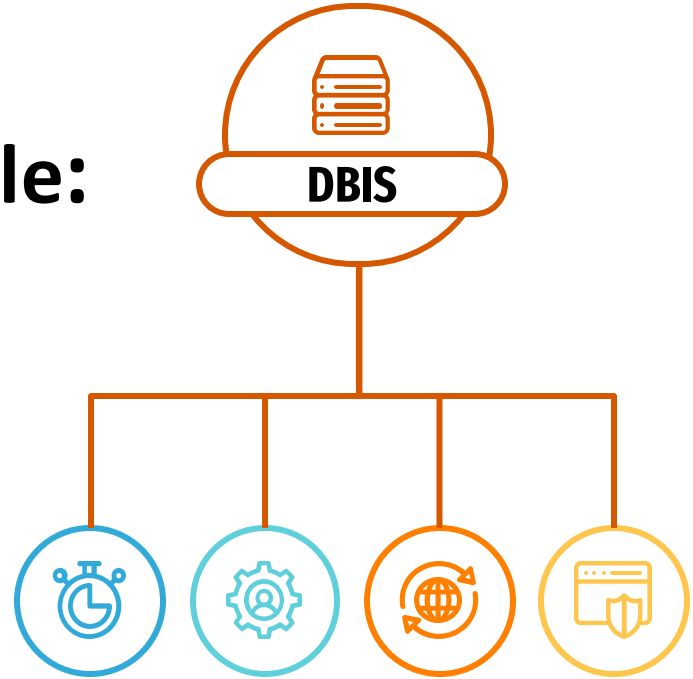
# Synchronisationskontrolle:

## 2-Phasen-Commit- vs. Paxos-Protokoll

Vergleich, Anwendungen und Unterschiede



# Agenda

1. Transaktionen in Verteilten Systemen
2. 2-Phasen-Commit-Protokoll
3. Paxos-Protokoll
4. Vergleich
5. Anwendungsbeispiele
6. Diskussion

# Lernziele

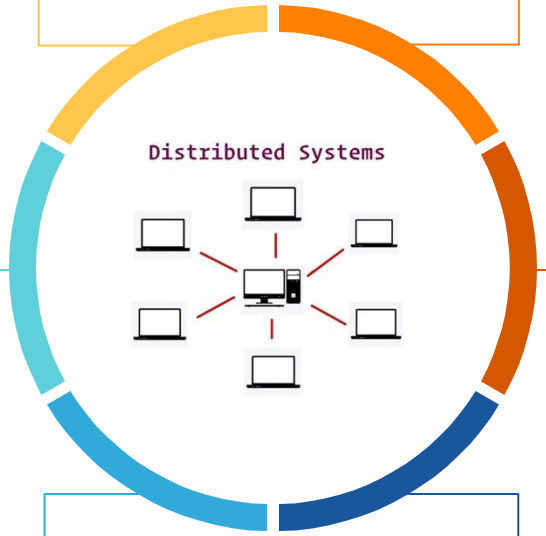
| | |
|---|---|
| Protokolle | Relevanz der beiden Konsensprotokolle im Kontext von verteilten Systeme verstehen |
| Vor- & Nachteile | Vorteile und Nachteile der Charakteristiken der Protokolle verstehen |
| Anwendungen | Vielfalt der Einsatzmöglichkeiten und Wichtigkeit im Alltag erkennen |
| Szenarienanalyse | Für bestimmte Szenarien die Eignung der Protokolle einordnen und Abläufe anwenden |

# Verteilte Systeme

**Mercury**
Mercury is the closest planet to the Sun

**Mars**
Despite being red, Mars is a cold place

**Jupiter**
It's a gas giant and the biggest planet




**Earth**
It's the planet where we all live on

**Venus**
Venus is the second planet from the Sun

**Saturn**
Saturn is the ringed one. It's a gas giant

# Isolation und Serialisierung

Grundsätzliche Forderung nach Konsistenz

- Beispiel: Überweisung
  - Von einem Konto bei Bank A wird etwas zu einem anderen Konto überwiesen
  - Kontostände müssen vor neuer Transaktion auf aktullstem Stand sein

# Isolation und Serialisierung

Grundsätzliche Forderung nach Konsistenz

**T1:** x = x + 1;
**T2:** x = x + 2;
**T3:** x = x + 3;

# Isolation und Serialisierung

Grundsätzliche Forderung nach Konsistenz

**T1:**
x = 0; (Initialzustand)
x = x + 1; (x wird 1)
Commit (x ist 1)

**T2:**
x = 1; (aus vorheriger Transaktion)
x = x + 2; (x wird 3)
Commit (x ist 3)

**T3:**
x = 3; (aus vorheriger Transaktion)
x = x + 3; (x wird 6)
Commit (x ist 6)

**→ x = 6**

# Isolation und Serialisierung

Grundsätzliche Forderung nach Konsistenz

**T1:** x = x + 1;
**T2:** x = x + 2;
**T3:** x = x + 3;

→ Race Conditions

# Isolation und Serialisierung

Grundsätzliche Forderung nach Konsistenz

**T1:** x = 0; (Initialzustand)
**T2:** x = 0;
**T1:** x = x + 1; (x wird 1)
**T3:** x = 0;
**T2:** x = x + 2; (x wird 3)
**T3:** x = x + 3; (x wird 6)
**T1:** Commit (x ist 1)
**T3:** Commit (x ist 3)
**T2:** Commit (x ist 2)

→ **x = 2**
→ **Lost update**

# 2-Phasen-Commit Protokoll

# Teilnehmer 2PC

1

**Zentraler Koordinator**

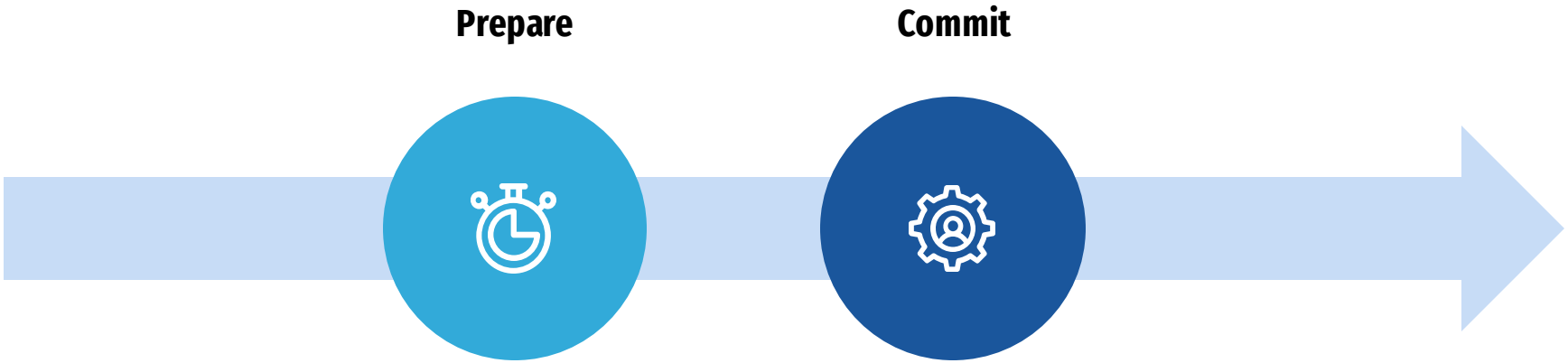
Schlägt eine Maßnahme vor

n

**Teilnehmer**

Alle müssen eine Antwort abgeben, nur dann wird die Aktion durchgeführt

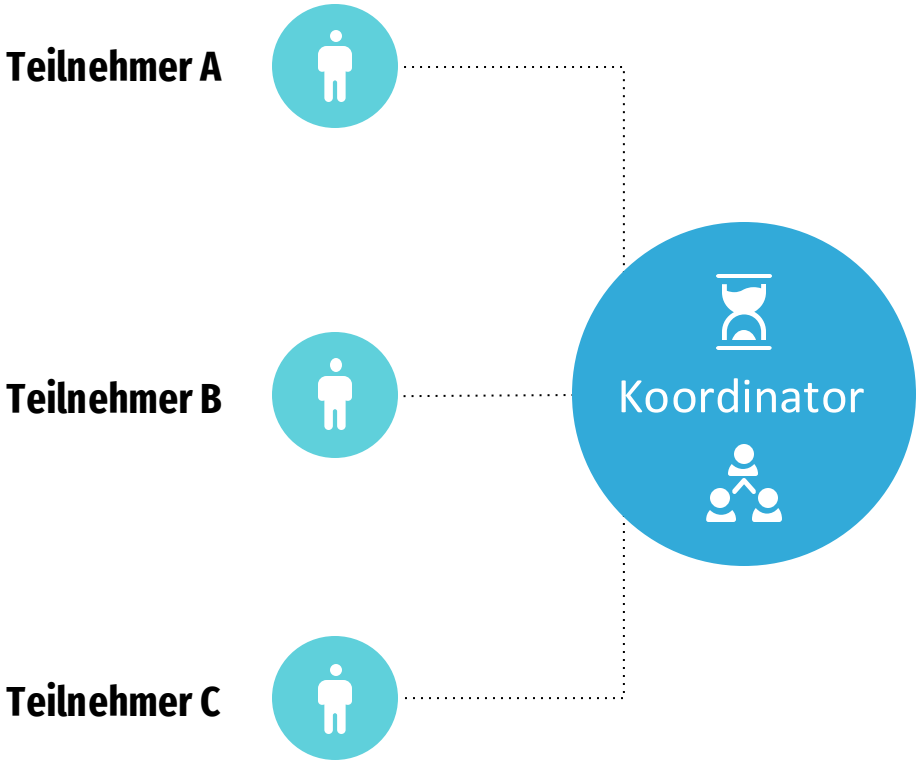
# 2PC Phasen



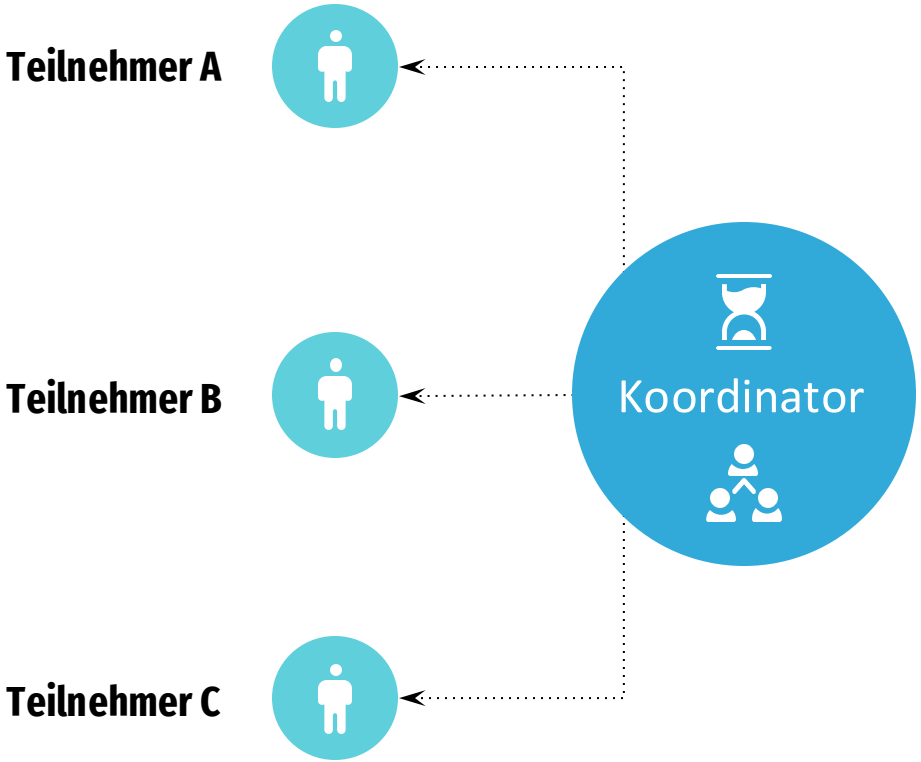

- Koordinator fragt nach Bereitschaft
- Komponenten melden Bereitschaft

- Koordniator prüft ob alle bereit sind
- Aktion wird (nicht) durchgeführt

# Ablauf 2PC



# Ablauf 2PC

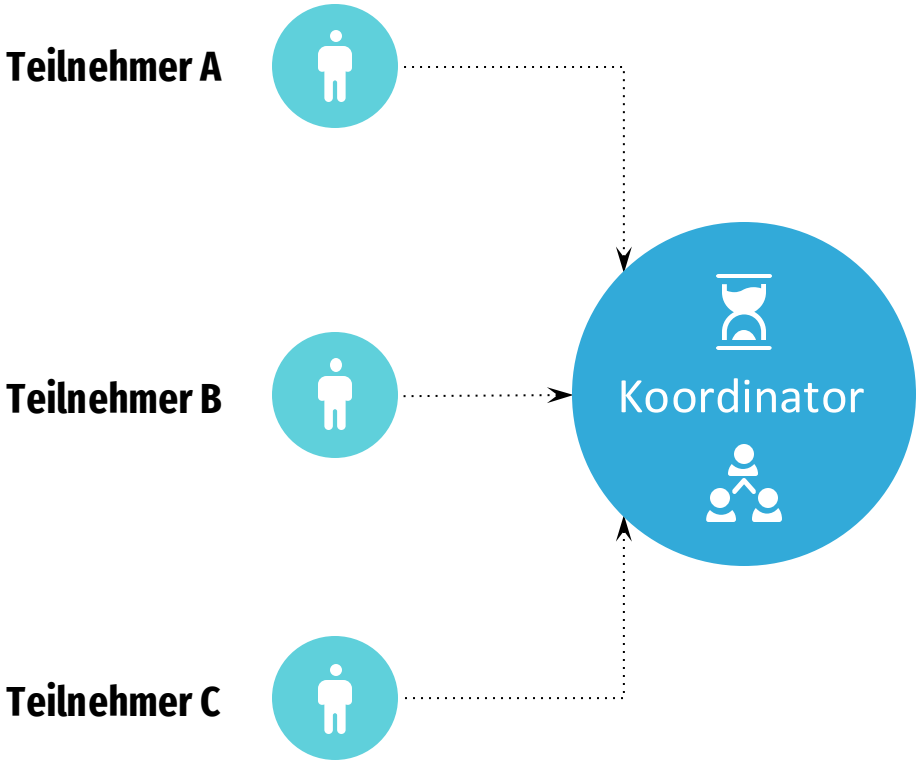



1

**Prepare**

Alle bereit?

# Ablauf 2PC

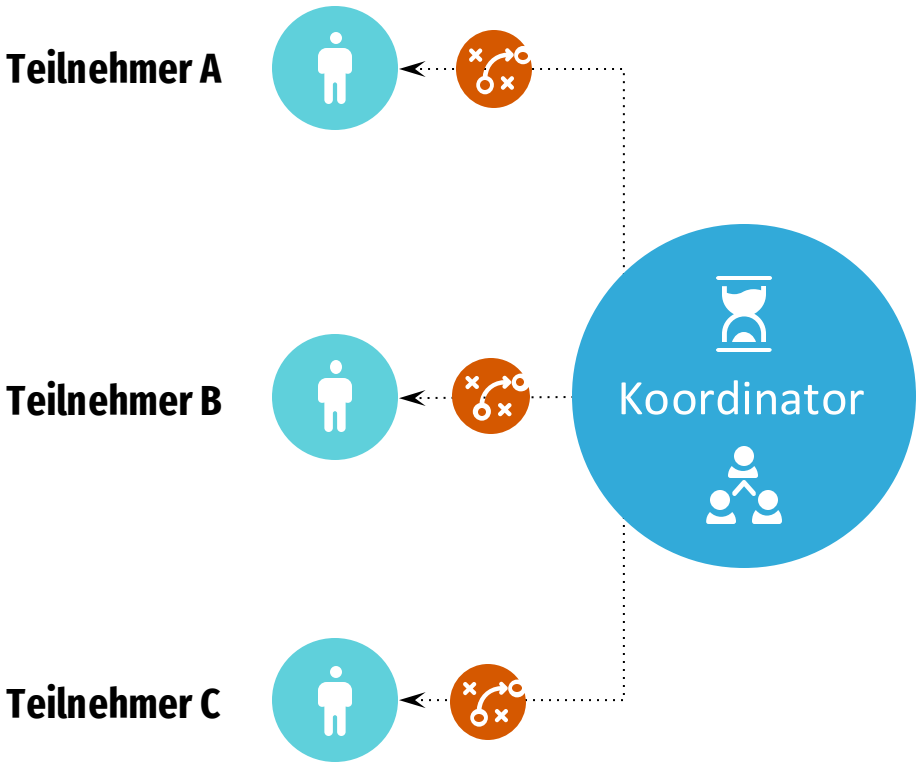



1

**Prepare**

- Ja, bereit
- Nein, nicht bereit

# Ablauf 2PC

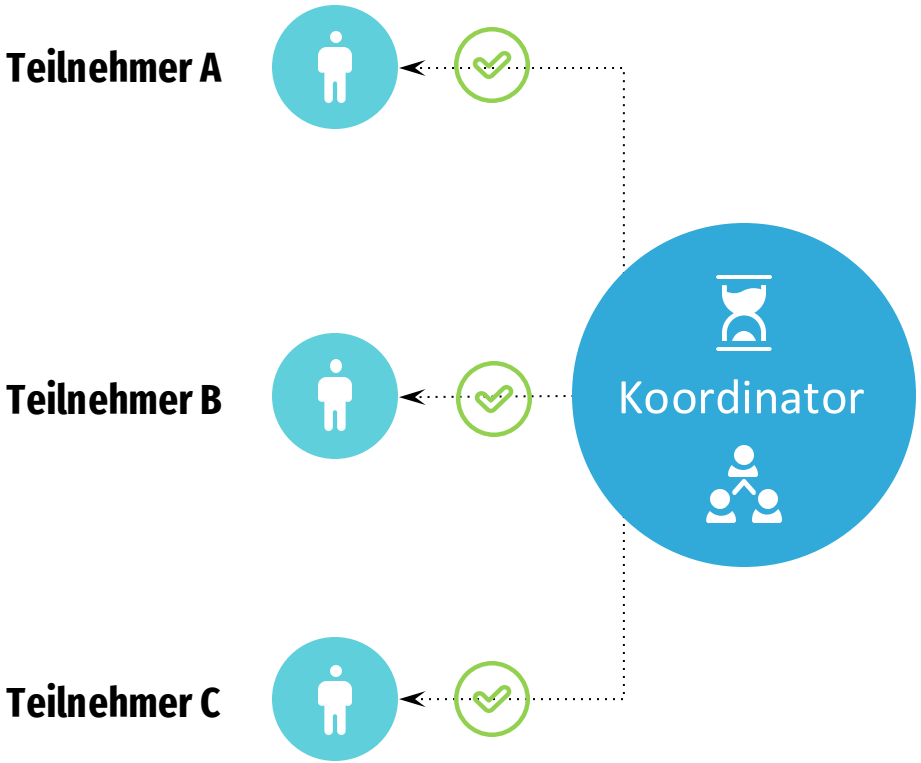



2

**Abort**

Mindestens einer sagt Nein

# Ablauf 2PC




2

**Commit**

Alle sagen Ja

# Paxos Protokoll

# Teilnehmer Paxos

1

**Proposer**

Initiiert einen Konsensprozess, kann irgendeiner der teilnehmenden Knoten sein. Dauert die Entscheidungsfindung zu lange, kann ein anderer Node übernehmen und zum Proposer warden.

n

**Acceptor**

Reagieren und beteiligen sich an Entscheidungsfindung

m

**Learner**

Können auch Instanzen sein, die nicht an der Entscheidungsfindung beteiligt waren

# Paxos Phasen

**Prepare**

- Ein Knoten nimmt Proposer-Rolle an
- Proposer macht Vorschlag

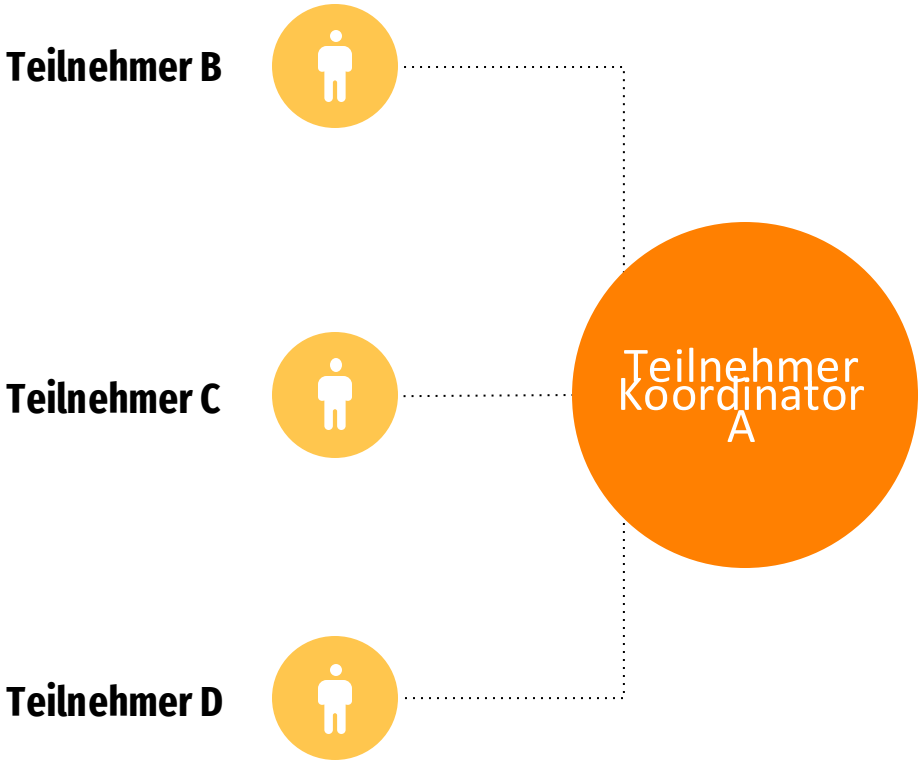
**Accept**

- Acceptors geben Feedback zu Vorschlag

**Commit/Learn**

- Wenn Mehrheit akzeptiert, dann wird ausgeführt

# Ablauf Paxos

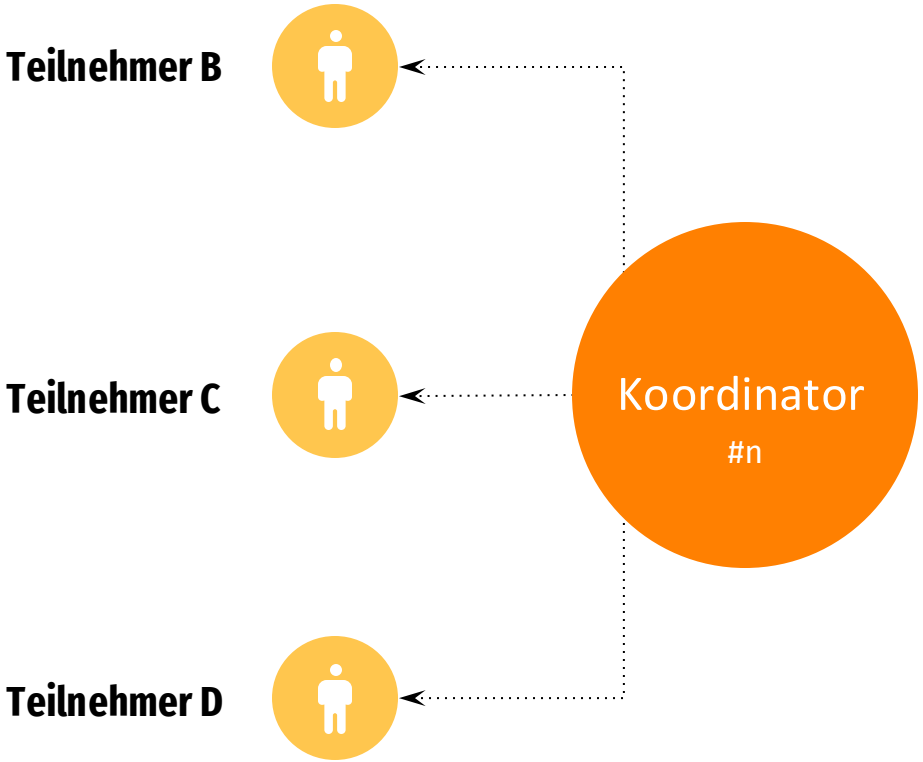



1

**Prepare**

Teilnehmer 1 initiiert und nimmt die Rolle vom Koordinator am

# Ablauf Paxos

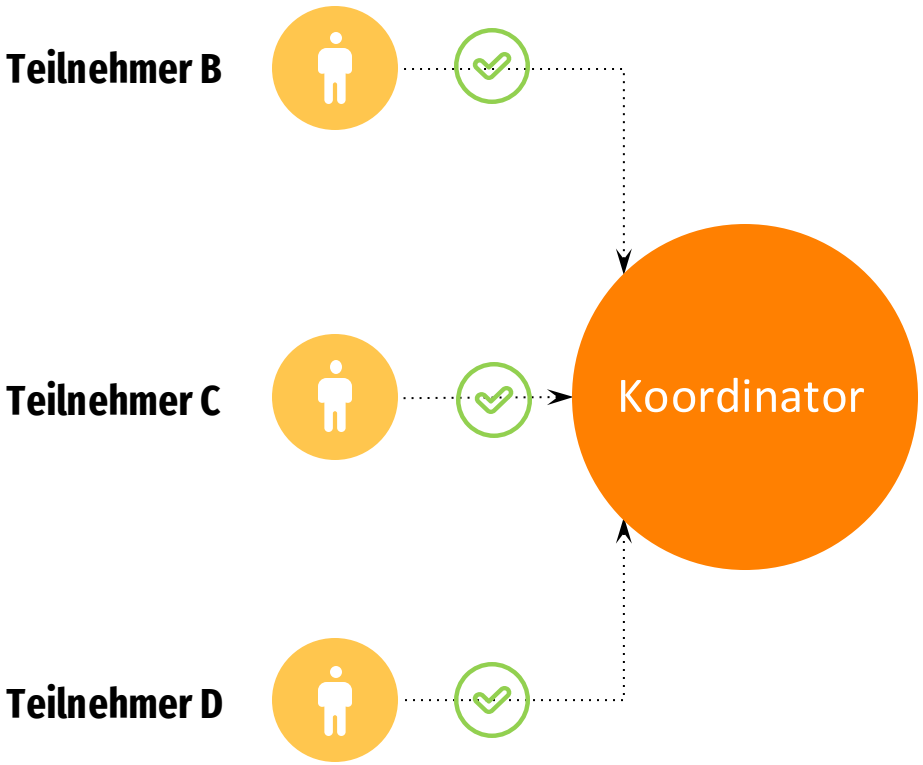



1

**Prepare**

Sendet Nachricht mit Proposal-ID an alle Teilnehmer

# Ablauf Paxos

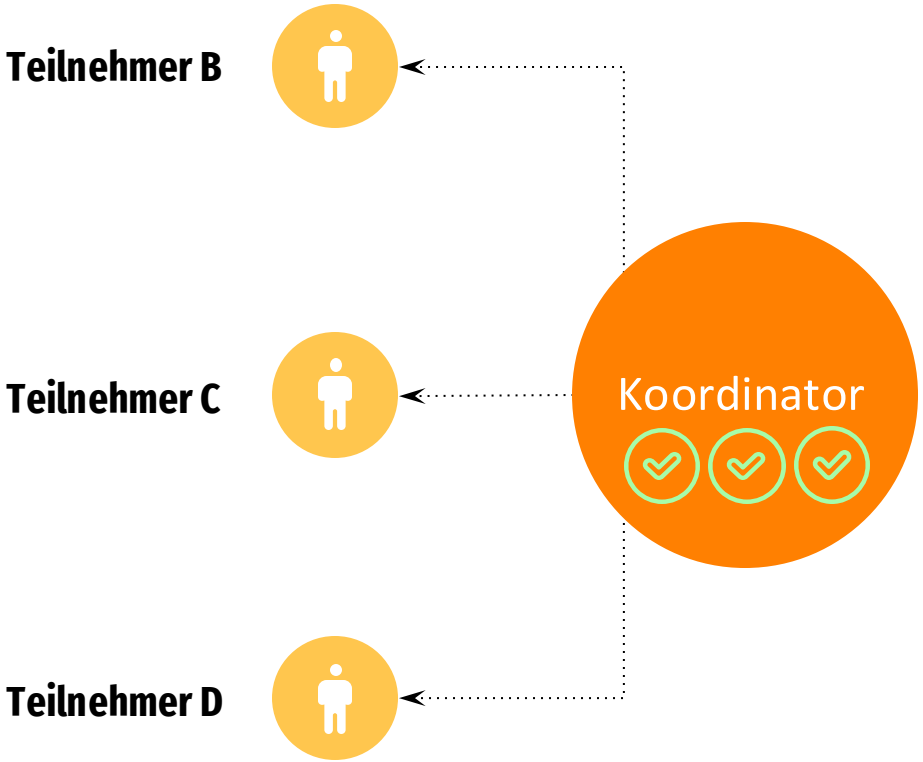



1

**Prepare**

Teilnehmer senden Promise zurück und versichern, dass dies die aktuellste Proposal-ID ist und sie bereit sind

# Ablauf Paxos

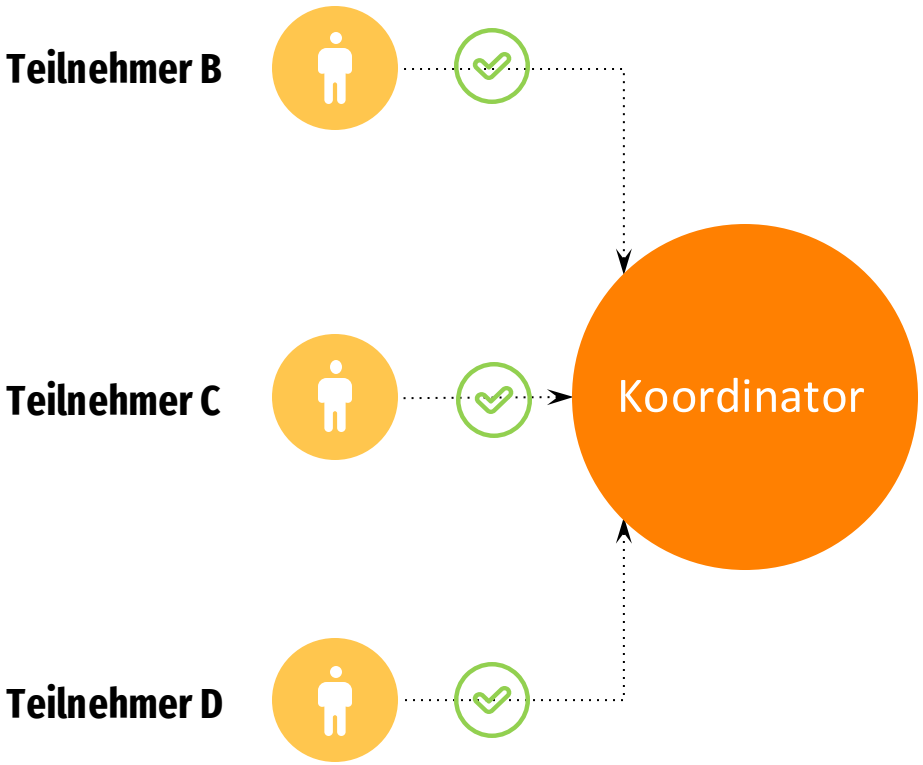



2

**Accept**

Sendet eine Accept-Nachricht mit Proposal-ID #n und Wert an Teilnehmer

# Ablauf Paxos

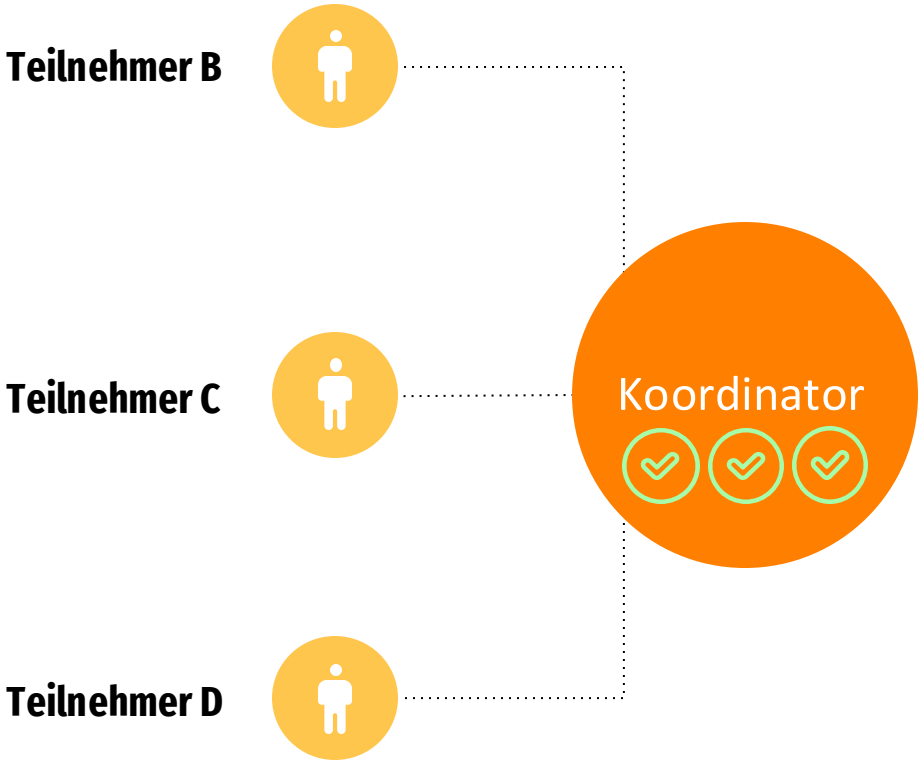



2

**Accept**

Teilnehmer bestätigen mit einer Accepted-Nachricht die Daten

# Ablauf Paxos

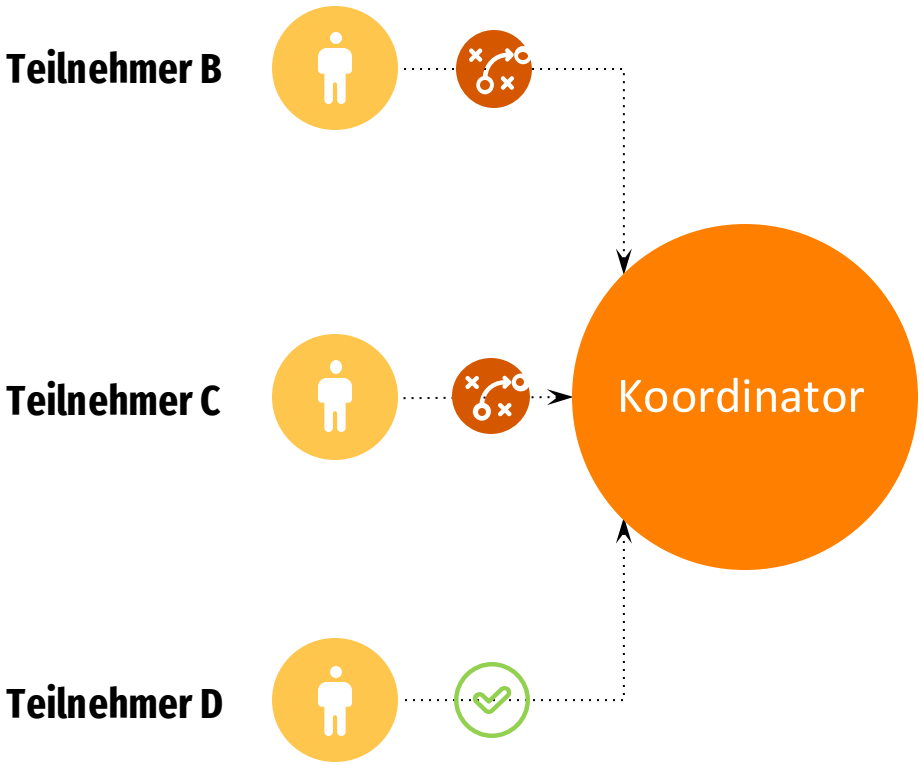



3

**Commit**

Koordinator empfängt Zustimmung von der Mehrheit für seinen Vorschlag und führt diesen aus

# Ablauf Paxos

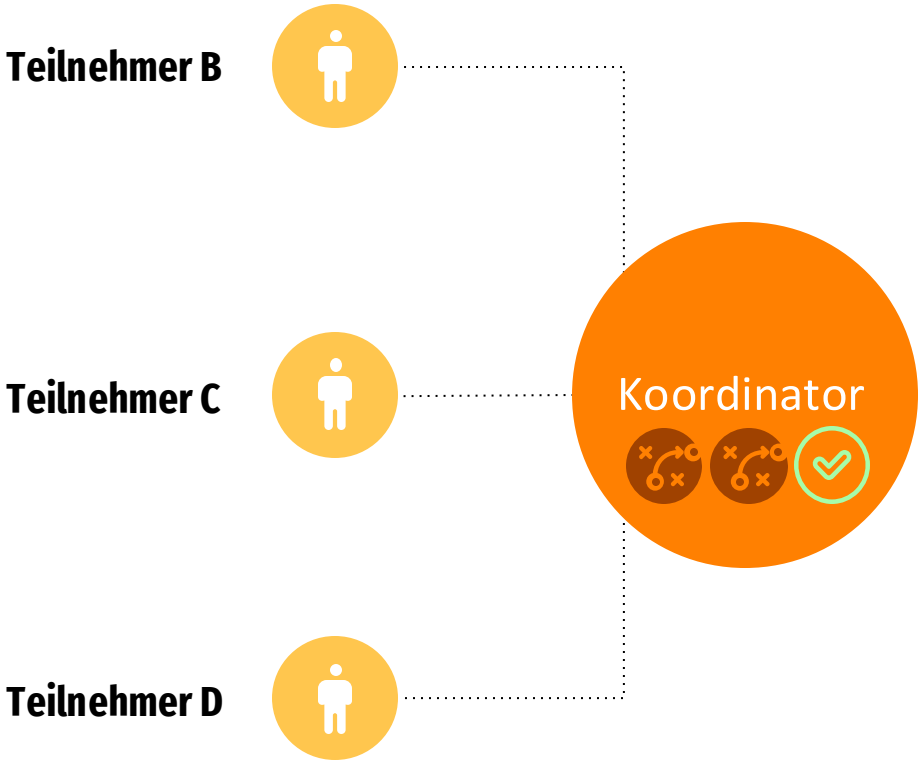



2

**Accept**

Zwei von drei Teilnehmern stimmen gegen den Vorschlag

# Ablauf Paxos

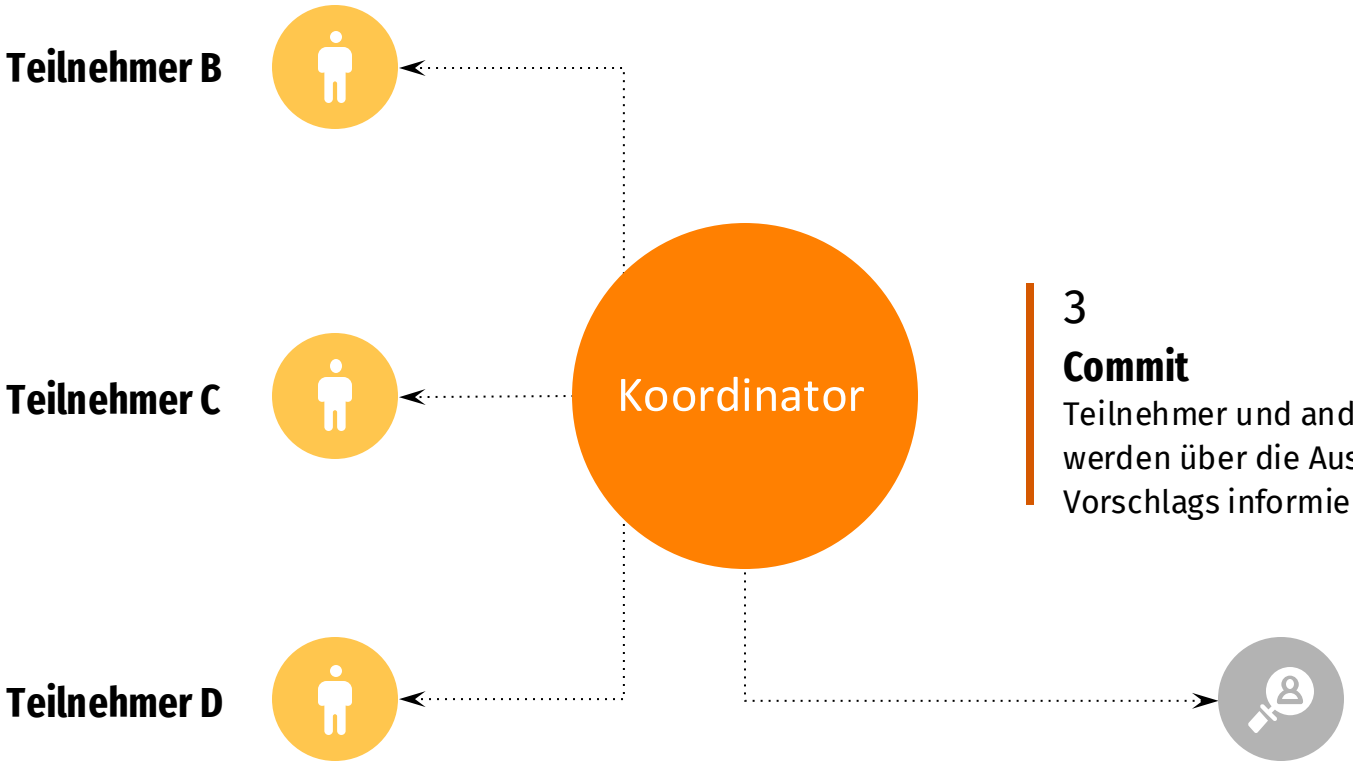



3

**Commit**

Koordinator empfängt keine Zustimmung von der Mehrheit und sendet im nächsten Schritt eine Abort-Nachricht (Abbruch der Transaktion)

# Ablauf Paxos




3

**Commit**

Teilnehmer und andere **Learner** werden über die Ausführung des Vorschlags informiert

# Lernziele

**Protokolle**

Relevanz der beiden Konsensprotokolle im Kontext von verteilten Systeme verstehen

# Charakteristik 2PC

**Flexible Steuerung**

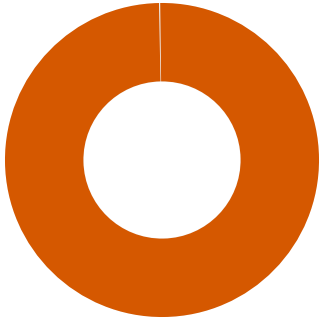
Je dezentraler, desto flexibler ist die Koordination

**Fehlertoleranz**

Wie gut kann das Protokoll mit Fehlern umgehen?

**Wiederholung**

Wiederaufnahme nach Systemausfall



**Implementation**

Simplizität der Implementation

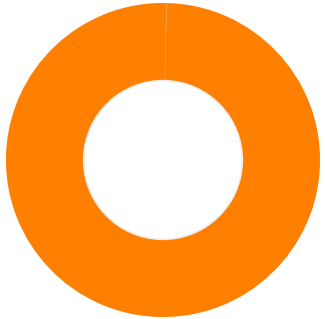
# Charakteristik Paxos

**Flexible Steuerung**

Je dezentraler, desto flexibler ist die Koordination

**Fehlertoleranz**

Wie gut kann das Protokoll mit Fehlern umgehen?



**Wiederholung**

Wiederaufnahme nach Systemausfall

**Implementation**

Simplizität der Implementation

# Vergleich

## 2-Phasen-Commit

**Simpel**

It’s a gas giant and the biggest planet in the Solar System

**Zentral**

Feste Rollen, zentraler Koordinator

## Paxos

**Sicher**

Ausfallsicherheit und Wiedeaufnahme-Möglichkeit

**Dezentral**

Flexible Rollen, Gegenvorschläge

# Lernziele

| | |
|---|---|
| Protokolle | Relevanz der beiden Konsensprotokolle im Kontext von verteilten Systeme verstehen |
| Vor- & Nachteile | Vorteile und Nachteile der Charakteristiken der Protokolle verstehen |

# Anwendungsbereiche

In welchen Bereichen sind Konsensprotokolle relevant?

(Branchen, Beispiele, etc.)

https://www.menti.com/alcp77wbnsas

Code: 4198 0680

# Anwendungsbeispiele

# Quiz

Welches Protokoll sollte für bestimmte Szenarien verwendet werden und warum?

https://www.menti.com/alcp77wbnsas

Code: 4198 0680

# Autonomer Verkehr

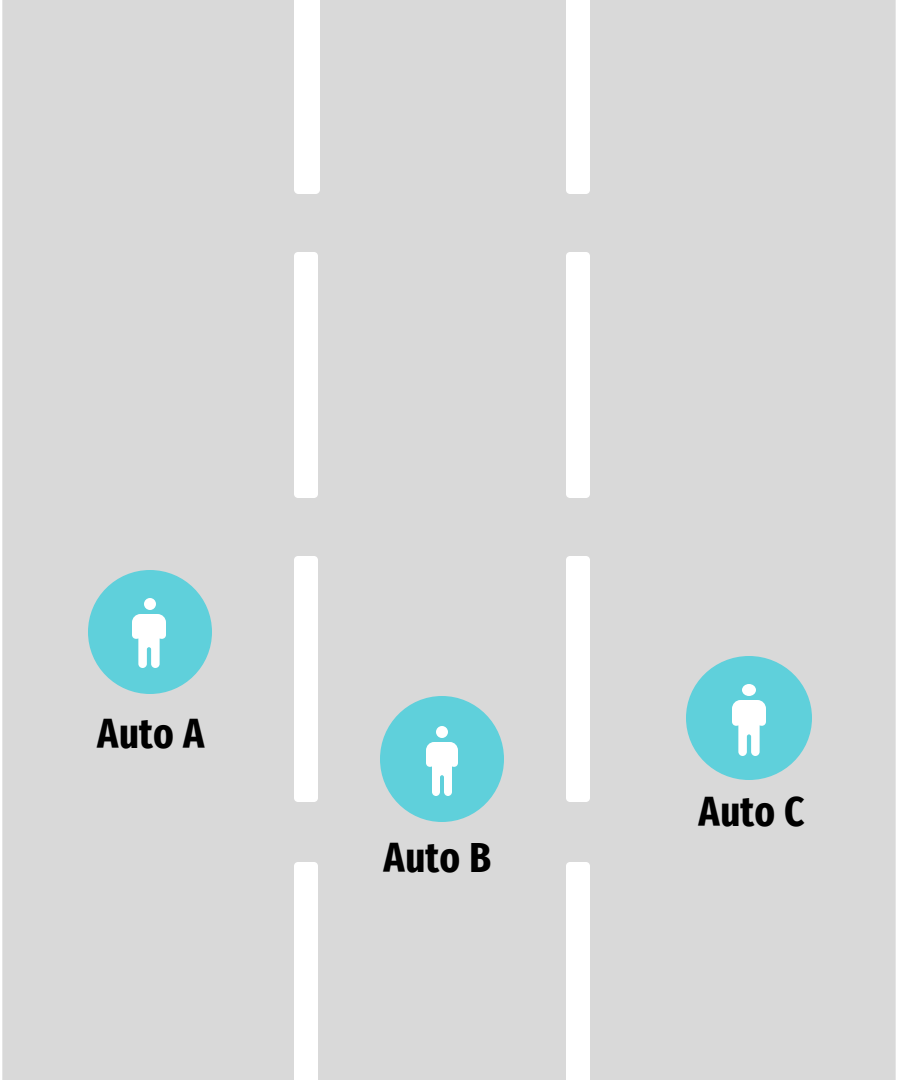
## Szenario

B erhält Promise-Nachrichten und
B wird aufgefordert, seinen Vorschlag
zu ändern
B sendet Prepare(1) → {A, C}
A und C passen sich ggf. entsprechend
an, um den sicheren Ablauf zu
gewährleisten

Damit senden sie Einverständnis, da
keine früheren Proposal-IDs
akzeptiert wurden

Bandenmodell gut geeignet ist,
weil das Fahrzeug seinen Kurs
ändern kann
Erfolg mit beiden anderen
Fahrzeugen kommunizieren
B generiert die Proposal-ID p=1
B sendet Accept(1, 120) → {A, C}
Da beide Sender die benötigte
Geschwindigkeit v=120
B beginnt auf 120 zu beschleunigen

A sendet Accepted(1, 120) → B
C sendet Accepted(1, 120) → B
A sendet Promise(1, null, null) → B
C sendet Promise(1, null, null) → B

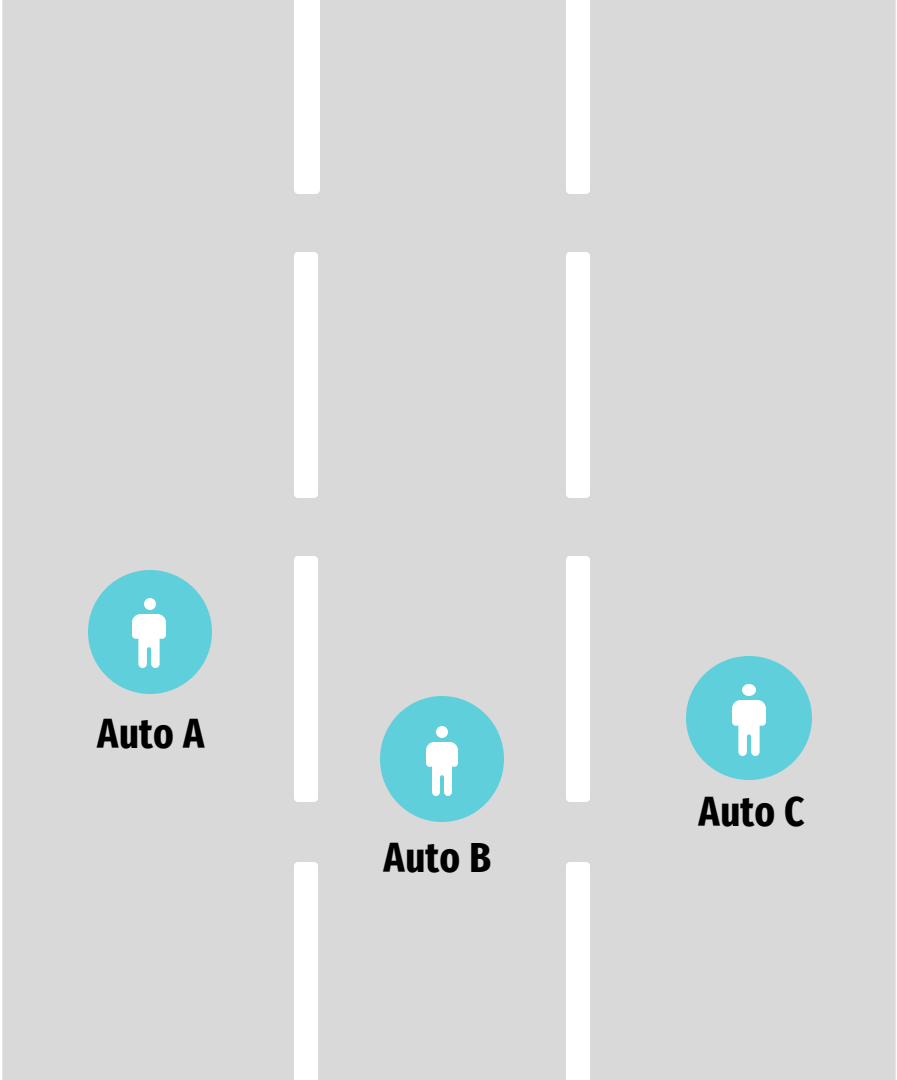


# Autonomer Verkehr

0

**Szenario**

B will auf die linke Spur wechseln
- Er muss mit beiden anderen Fahrzeugen kommunizieren
- A darf nicht schneller werden
- C darf nicht auch in diesem Moment überholen wollen



# Autonomer Verkehr

1

**Prepare**

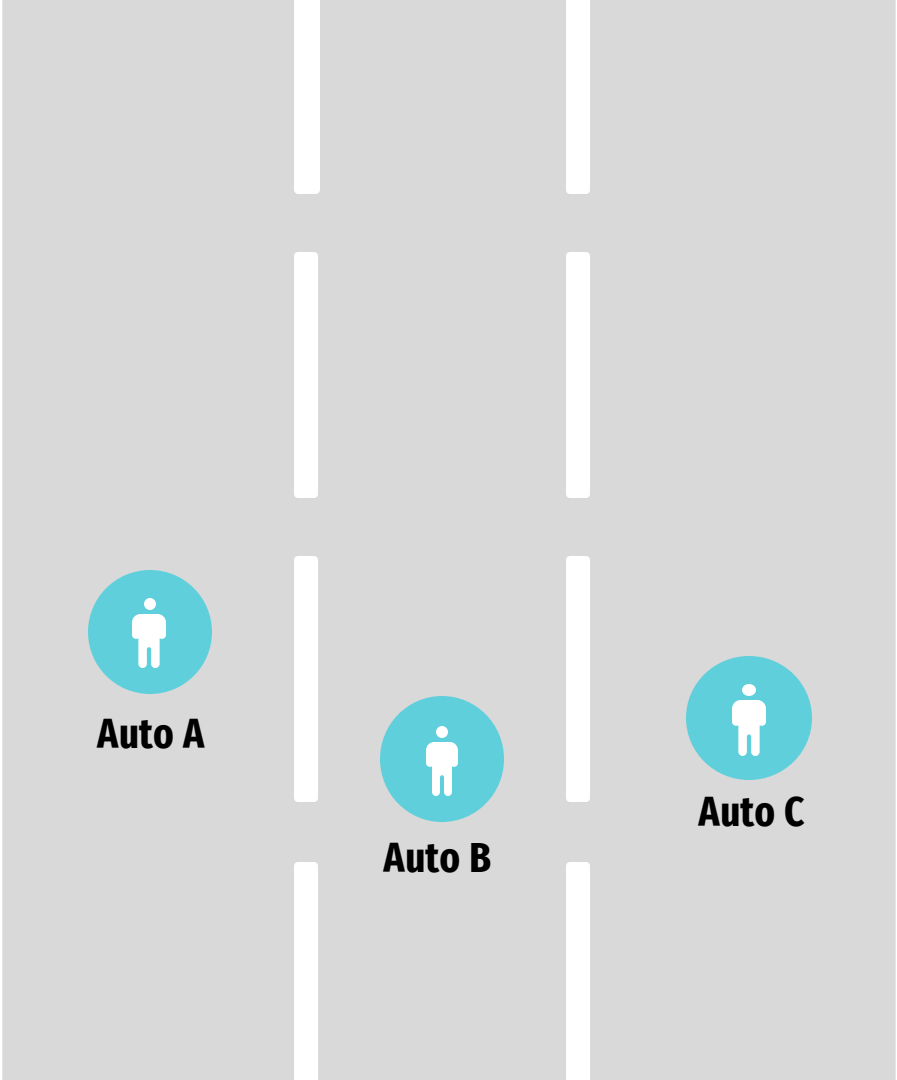
B nimmt Koordinator-Rolle an und initiiert einen Vorschlag

Er generiert die Proposal-ID p=1
Er bestimmt die benötigte Geschwindigkeit v=120

B sendet Prepare(1) → {A, C}

A sendet Promise(1, null, null) → B
C sendet Promise(1, null, null) → B

Damit senden sie Einverständnis, da keine früheren Proposal-IDs akzeptiert wurden



# Autonomer Verkehr

2

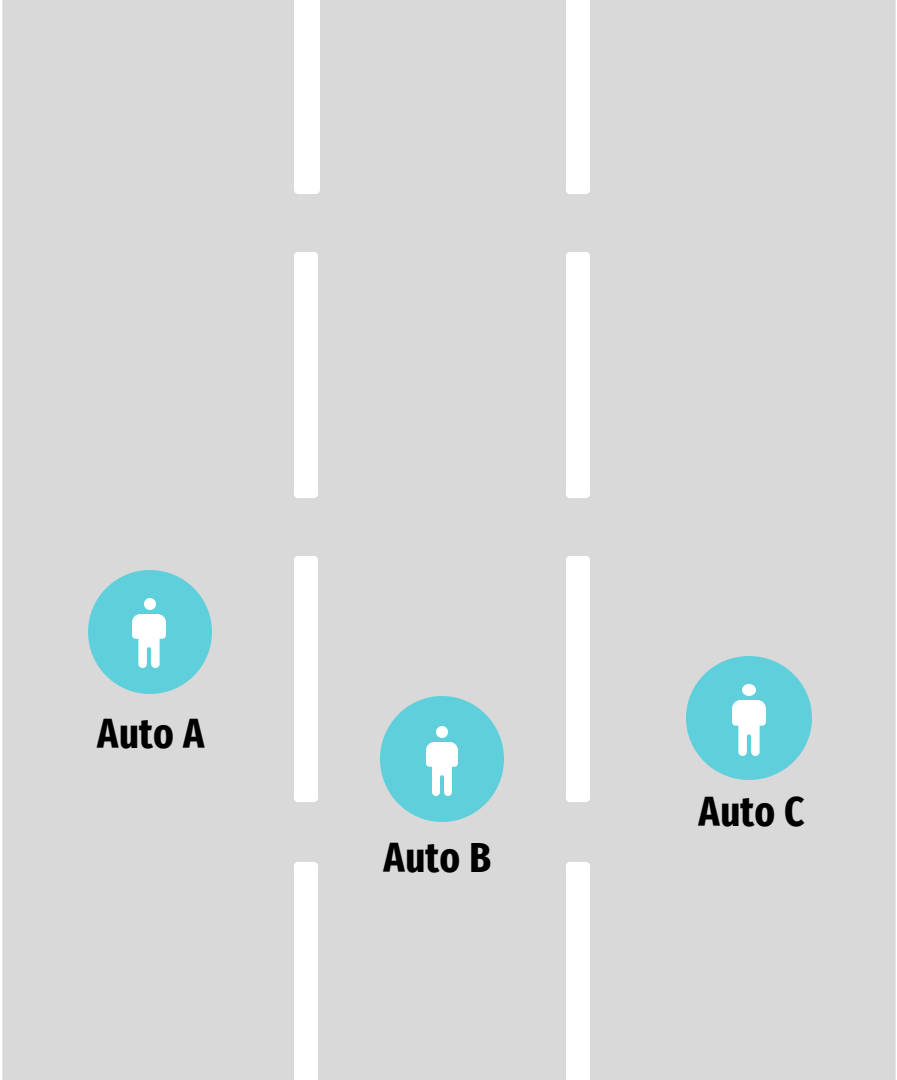
**Accept**

B erhält Promise-Nachrichten und weiß, dass er seinen Vorschlag senden darf

B sendet Accept(1, 120) → {A, C}

A und C prüfen, ob dies machbar ist und senden ihr Feedback zurück

A sendet Accepted(1, 120) → B
C sendet Accepted(1, 120) → B



# Autonomer Verkehr

3

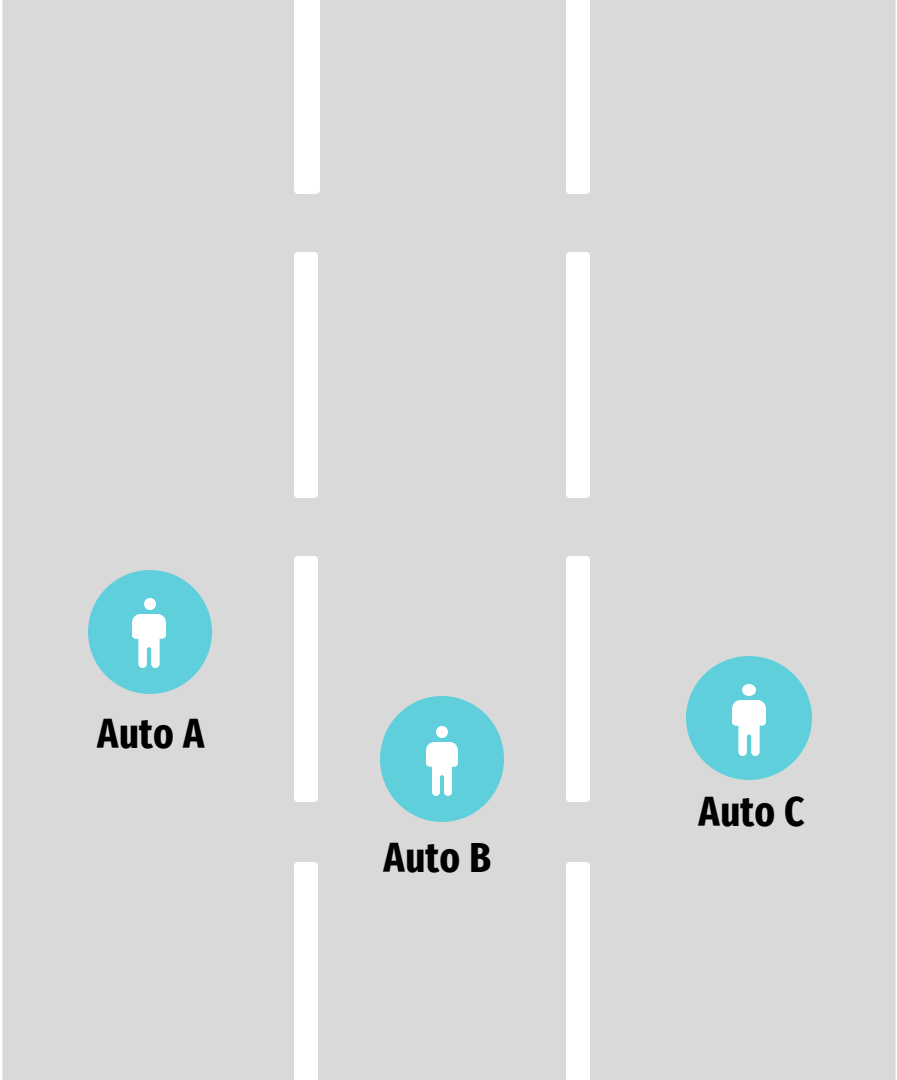
**Commit**

Da die Mehrheit dafür gestimmt hat, kann der Vorschlag von B durchgeführt werden

Da die B sendet Commit(1, 120) → {A, C}

B beginnt auf 120 zu beschleunigen

A und C passen sich ggf. entsprechend an, um den sicheren Ablauf zu gewährleisten



# Lernziele

| | |
|---|---|
| Protokolle | Relevanz der beiden Konsensprotokolle im Kontext von verteilten Systeme verstehen |
| Vor- & Nachteile | Vorteile und Nachteile der Charakteristiken der Protokolle verstehen |
| Anwendungen | Vielfalt der Einsatzmöglichkeiten und Wichtigkeit im Alltag erkennen |

# Übung

## Pizza?

Sechs Freunde (A, B, C, D, E) planen ein gemeinsames Abendessen.

- A schlägt vor, Pizza zu machen.
- B hat kein Internet und erhält die Nachricht nicht.
- Die anderen sind online
- C und D mögen Pizza nicht

1. Welches Protokoll würden Sie zur Entscheidungsfindung eher verwenden und warum?
2. Notiere die Schritte und welche Entscheidung am Ende getroffen wird.
3. Was wäre anders, wenn B mit abgestimmt hätte und auch *für* Pizza gestimmt hätte?

# Lernziele

| | |
|---|---|
| Protokolle | Relevanz der beiden Konsensprotokolle im Kontext von verteilten Systeme verstehen |
| Vor- & Nachteile | Vorteile und Nachteile der Charakteristiken der Protokolle verstehen |
| Anwendungen | Vielfalt der Einsatzmöglichkeiten und Wichtigkeit im Alltag erkennen |
| Szenarienanalyse | Für bestimmte Szenarien die Eignung der Protokolle einordnen und Abläufe anwenden |

# Referenzen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- https://www.mydistributed.systems/2021/04/paxos.html (pseudo code)

- 2PC
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- Bernstein, P. A., & Newcomer, E. (2009). *Principles of Transaction Processing*. Morgan Kaufmann.
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- Verteilte Systeme
- **https://www.ifi.tu-clausthal.de/fileadmin/IFI/Abteilungen/Energieinformationsnetze_und_Systeme/VerteilteSys_Skript-1_02.pdf (auch 2pc gut beschrieben S.91-96)**
- https://www.sangam.dev/blog/2022/distributed-systems-key-concepts-and-challenges (Bild)

# Ablauf 2PC

**Teilnehmer A**

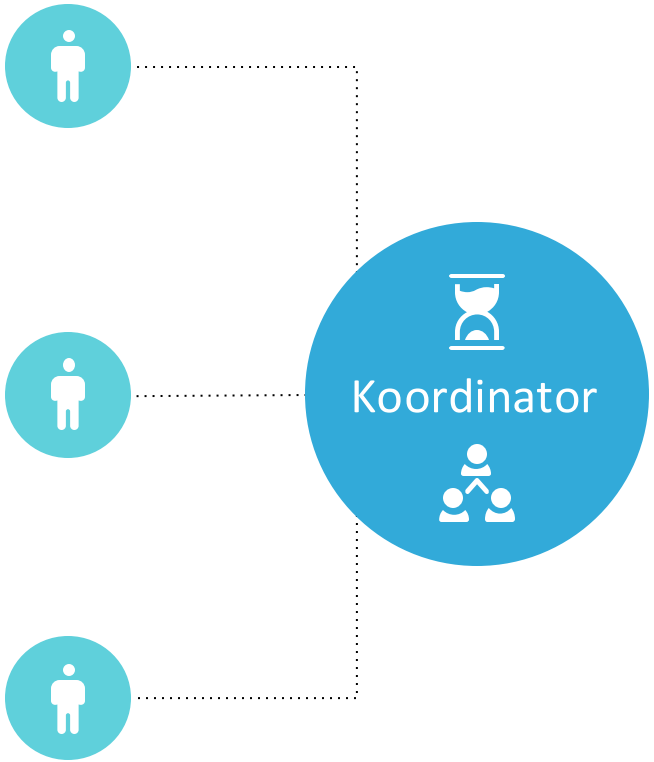
Mercury is the closest planet to the Sun

**Teilnehmer B**

Despite being red, Mars is a cold place

**Teilnehmer C**

Earth is the planet where we live on



# Ablauf Paxos

**Teilnehmer A**

Mercury is the closest planet to the Sun

**Teilnehmer B**

Despite being red, Mars is a cold place

**Teilnehmer C**

Earth is the planet where we live on

# Big Data Infographics

Here is where this template begins

# Big Data Infographics

**46%**

Mercury is the closest planet to the Sun

**34%**

Despite being red, Mars is a cold place

**22%**

Neptune is the farthest planet from the Sun

**68%**

Jupiter is the biggest planet of them all

### Mercury

Mercury is the closest planet to the Sun

### Mars

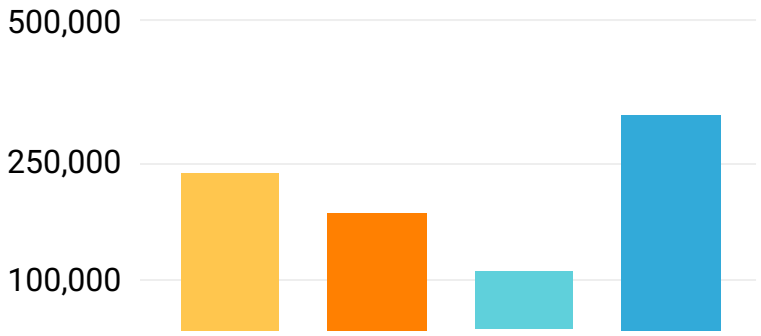
Despite being red, Mars is a cold place

### Neptune

Neptune is the farthest planet from the Sun

### Jupiter

It's the biggest planet in the Solar System



# Big Data Infographics

### Reduce costs

Venus has a beautiful name and is the second one from the Sun

### Improve productivity

Mercury is the closest planet to the Sun and the smallest one

### Attract new customers

Despite being red, Mars is a cold place full of iron oxide dust

### New facilities/Geo expansion

Jupiter is a gas giant and the biggest planet in the Solar System

# Big Data Infographics

**Mercury**
Mercury is the closest planet to the Sun

**Mars**
Despite being red, Mars is a cold place

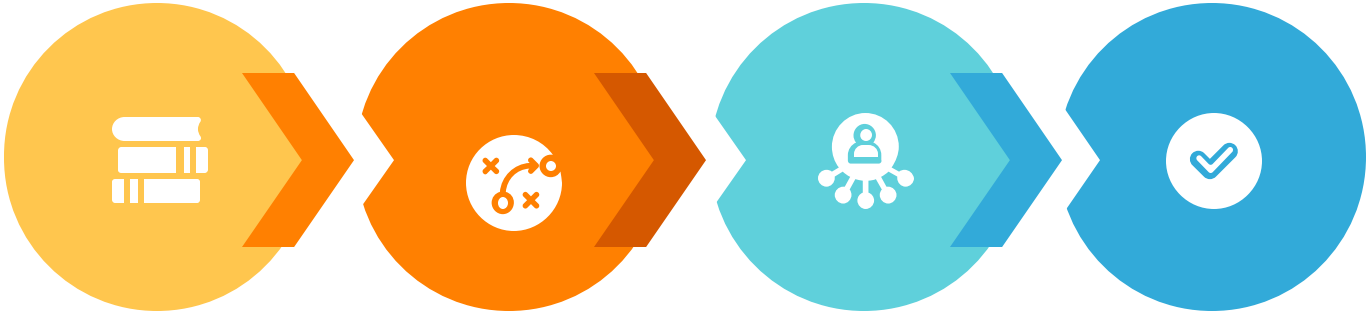
**Jupiter**
It's a gas giant and the biggest planet

**Earth**
It's the planet where we all live on

**Venus**
Venus is the second planet from the Sun

**Saturn**
Saturn is the ringed one. It's a gas giant

# Big Data Infographics



**Educate**

Mercury is the closest planet to the Sun and the smallest one

**Explore**

Despite being red, Mars is actually a cold place. It’s full of iron oxide dust

**Engage**

Jupiter is a gas giant and the biggest planet in the Solar System

**Execute**

Mercury is the closest planet to the Sun and the smallest one

# Big Data Infographics

**75%**

**Mars**

Mars is actually actually a cold place. It’s full of iron oxide dust

**60%**

**Saturn**

Saturn is composed of hydrogen and helium and is the ringed one

**50%**

**Mercury**

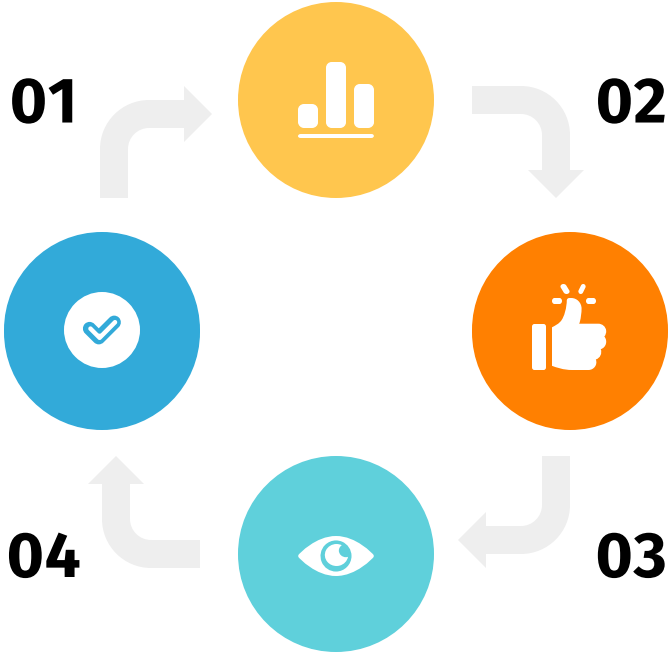
Mercury is the closest planet to the Sun and the smallest one

# Big Data Infographics

## Data

Despite being red, Mars is a cold place full of iron oxide dust

01



02

## Decision

Venus has a beautiful name, but also very high temperatures

## Info

Mercury is the closest planet to the Sun and the smallest one

04

03

## Insight

Jupiter is a gas giant and the biggest planet in the Solar System

# Big Data Infographics

### Data

Despite being red, Mars is a cold place. It's full of iron oxide dust

### Insight

Mercury is the closest planet to the Sun and the smallest one of them all

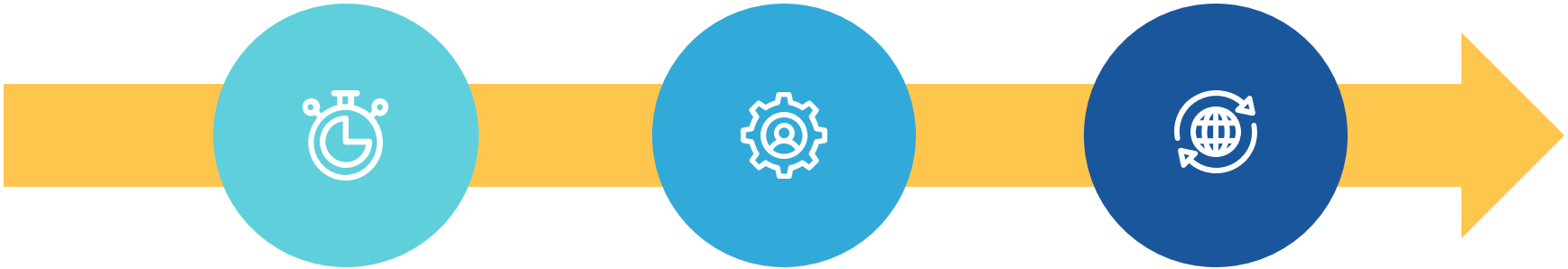
### Decision

Mars is full of iron oxide dust, which gives the planet its reddish cast

### Info

Jupiter is a gas giant and the biggest planet in the Solar System

# Big Data Infographics



**Mars**

Despite being red, Mars is actually a cold place. It’s full of iron oxide dust

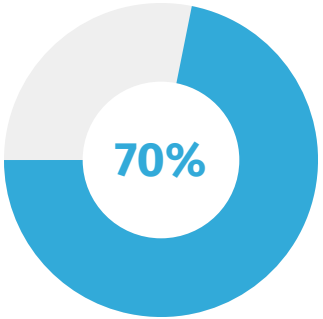
**Jupiter**

Jupiter is a gas giant and the biggest planet in the Solar System

**Mercury**

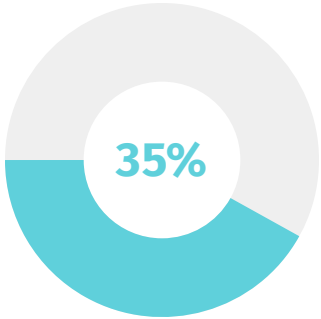
Mercury is the closest planet to the Sun and the smallest one

# Big Data Infographics



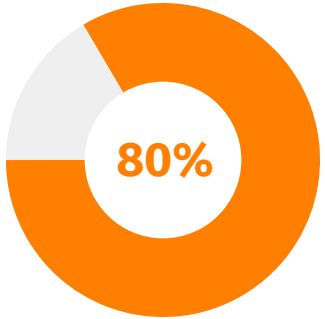

### Mercury

Mercury is the closest planet to the Sun and the smallest one



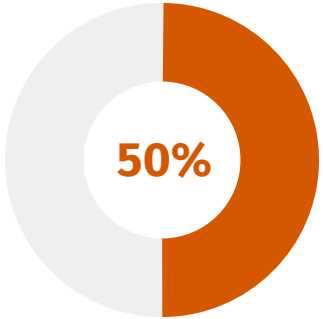

### Mars

Despite being red, is actually cold and full of iron oxide dust




### Jupiter

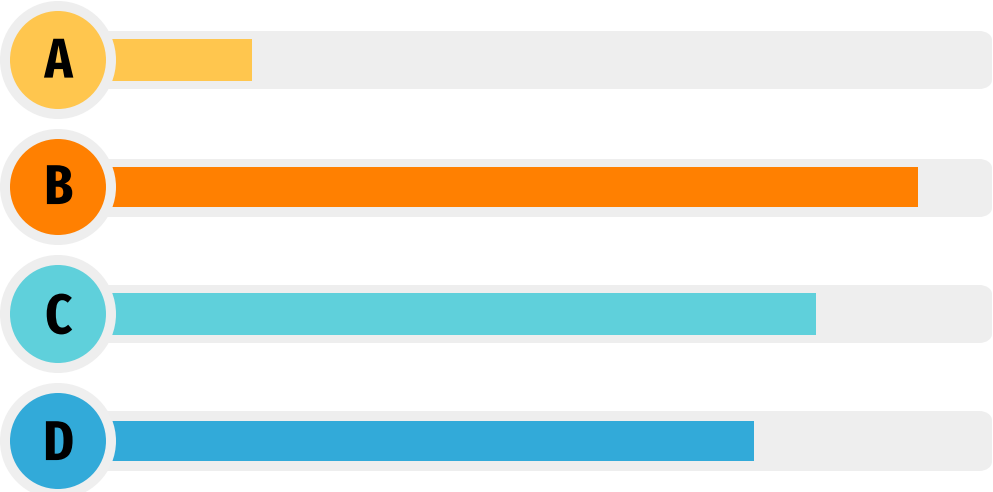
It’s a gas giant and the biggest planet in the Solar System




### Saturn

Saturn is composed mostly of hydrogen and also helium

# Big Data Infographics

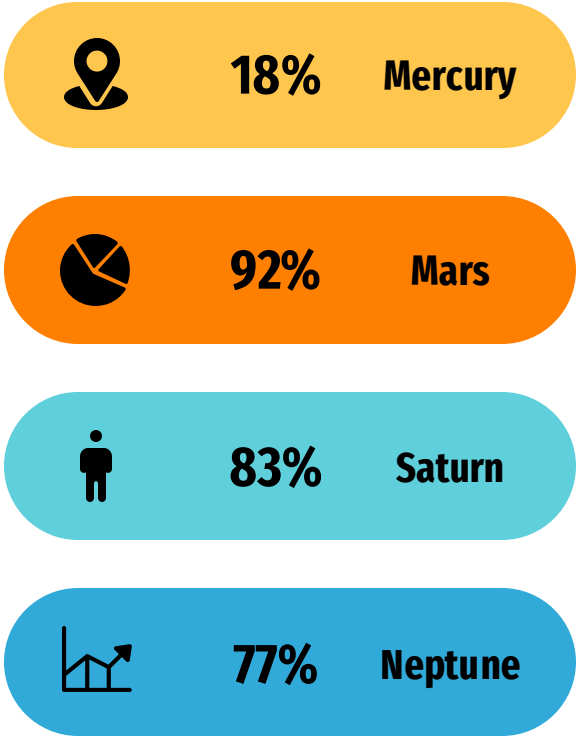



**Mercury**
Mercury is the closest planet to the Sun

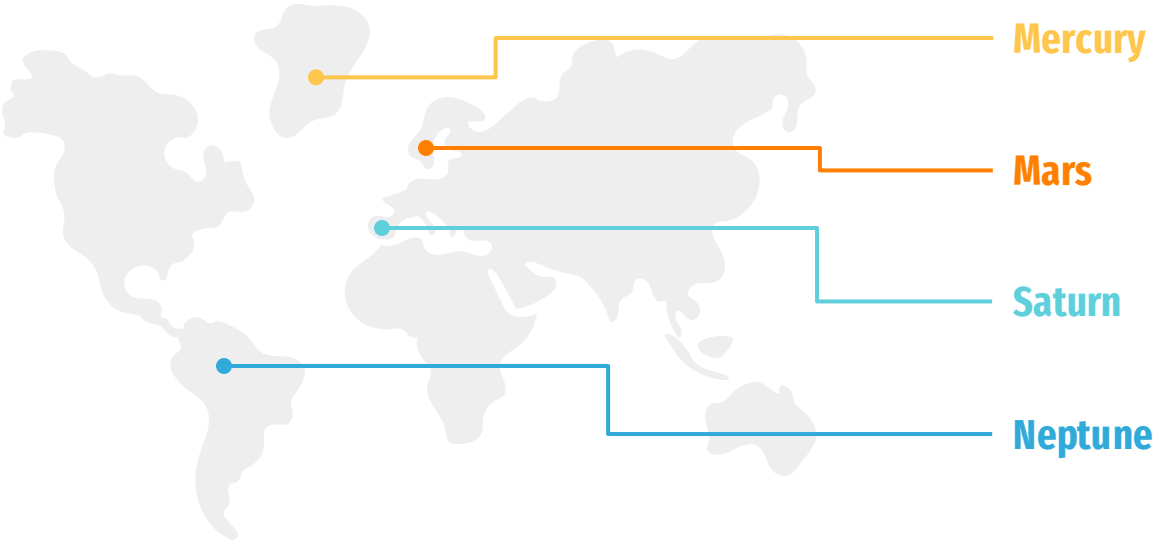
**Saturn**
It's composed of hydrogen and of helium

**Mars**
Despite being red, Mars is a cold place

**Neptune**
Neptune is the farthest planet from the Sun



# Big Data Infographics




Mercury is the closest planet to the Sun

Despite being red, Mars is a cold place

It’s composed of hydrogen and helium

It’s the farthest planet from the Sun

# Big Data Infographics

## Data Risk

### Data Security

Mercury is the closest planet to the Sun and the smallest

### Data Quality

Despite being red, Mars is cold and full of iron oxide dust

## Interpretation Risk

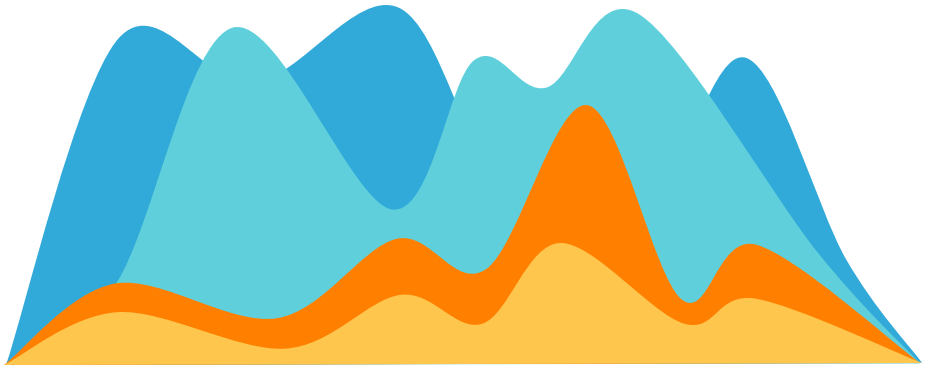
### Model Risk

It's a gas giant and the biggest planet in the Solar System

### Interpretation Risk

Saturn is composed mostly of hydrogen and helium

# Big Data Infographics



**Venus**



**Mercury**

**Mars**

**Jupiter**

**20%**

Venus has a beautiful name and is the second one from the Sun

**40%**

Mercury is the closest planet to the Sun and the smallest one

**60%**

Despite being red, Mars is a cold place full of iron oxide dust

**80%**

It’s a gas giant and the biggest planet in the Solar System

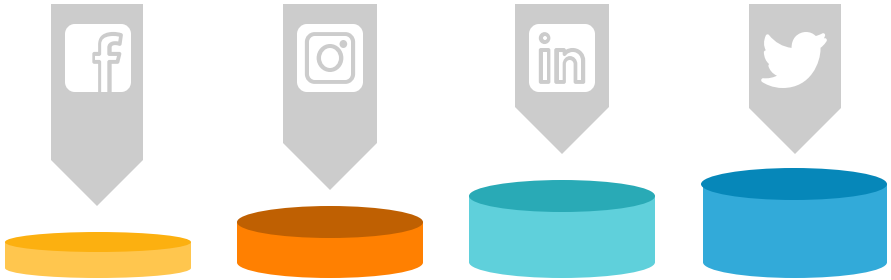
# Big Data Infographics

## Mercury

Mercury is the closest planet to the Sun

## Mars

Despite being red, Mars is a cold place

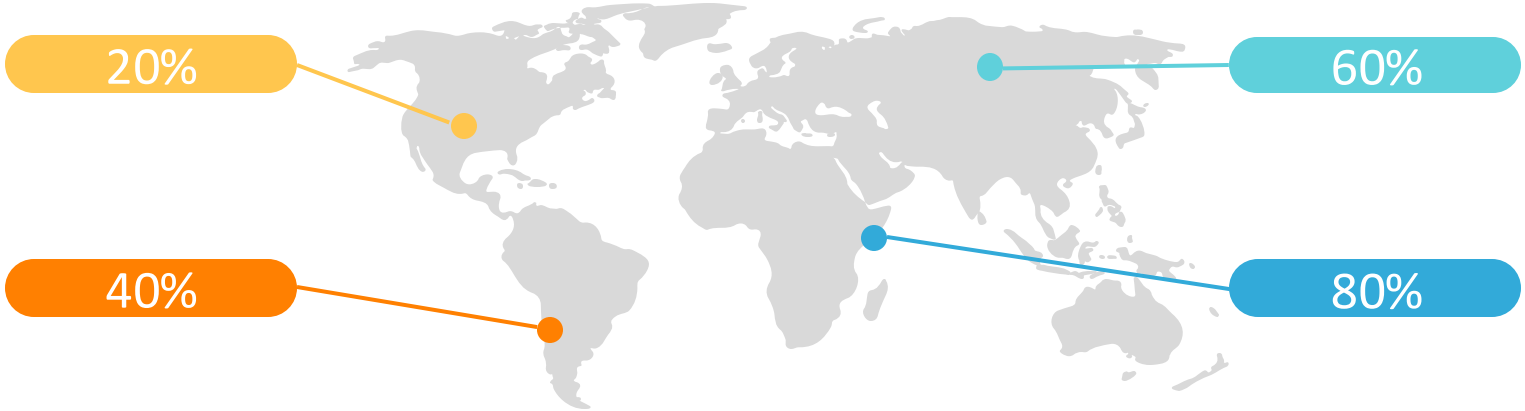


## Saturn

It's composed of hydrogen and helium

## Neptune

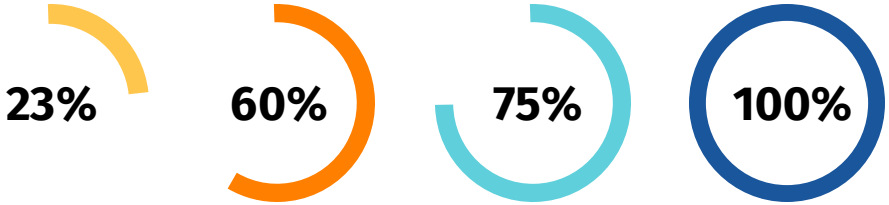
Neptune is the farthest planet from the Sun



# Big Data Infographics

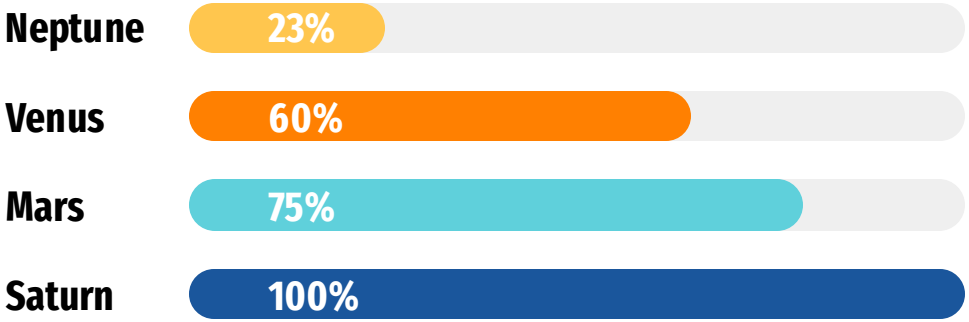
**Data**

Saturn is a gas giant and has several rings




**Decision**

Mars is full of iron oxide dust and has a reddish cast




**Insight**

Mercury is the smallest planet in the Solar System

**Info**

Jupiter is a gas giant and the biggest planet

# Big Data Infographics

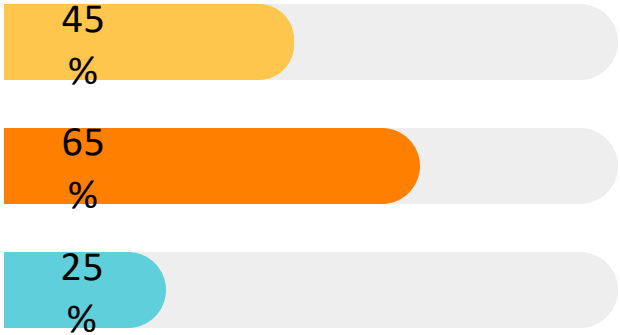
# Big Data Infographics

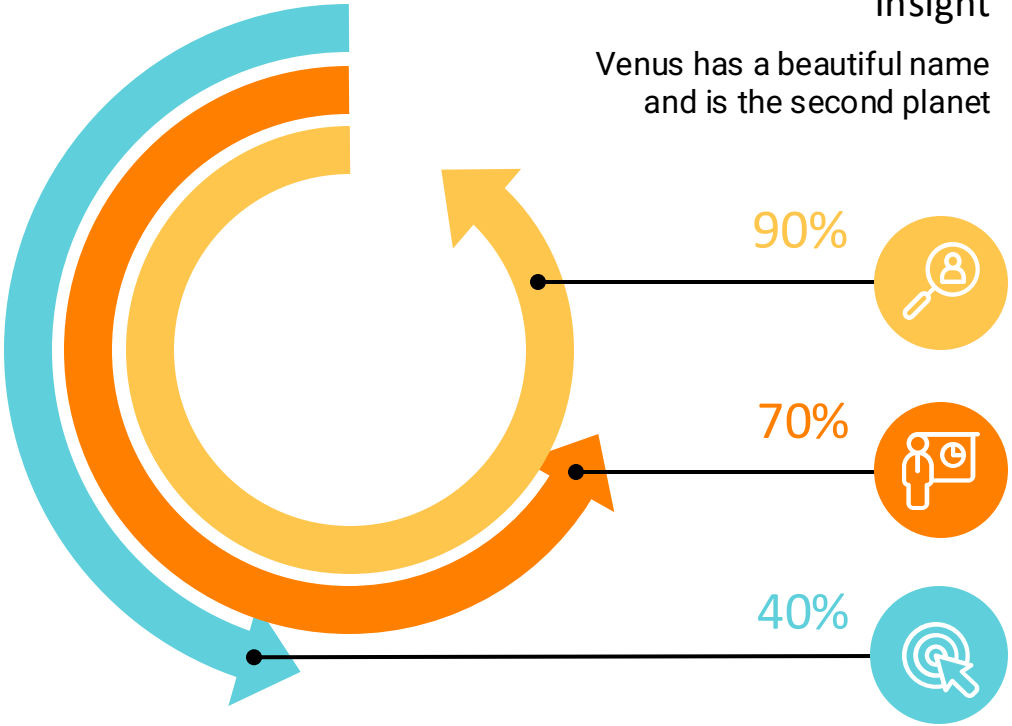
| | | |
|---|---|---|
| Data Security | Despite being red, Mars is actually a cold place | 60% |
| Data Quality | Mercury is the closest planet to the Sun | 45% |
| Model Risk | Jupiter is a gas giant and the biggest planet | 70% |
| Interpretation Risk | Yes, Saturn is the ringed one. It's a gas giant | 50% |

# Big Data Infographics

## Data

Despite being red, Mars is actually a very cold place






## Insight

Venus has a beautiful name and is the second planet

# Big Data Infographics

## Data interpretation

Mars is actually a cold place

## Structured data

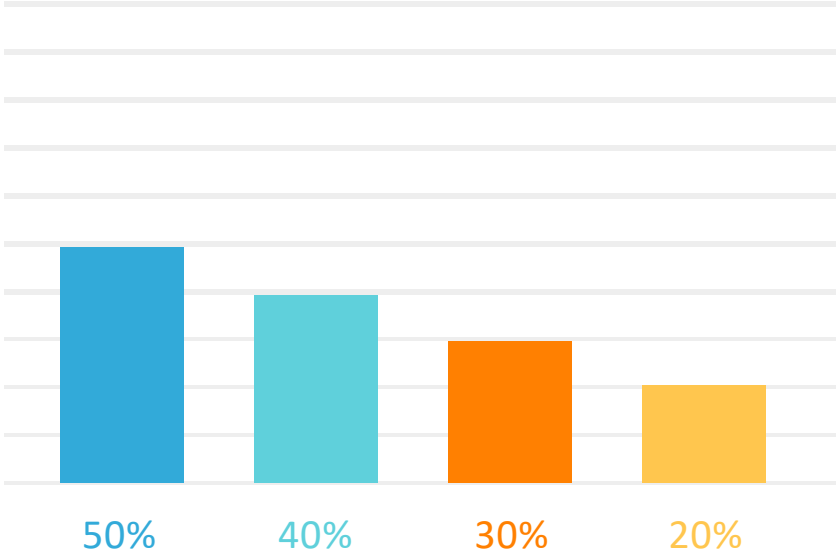
Jupiter is the biggest planet

## Geo expansion

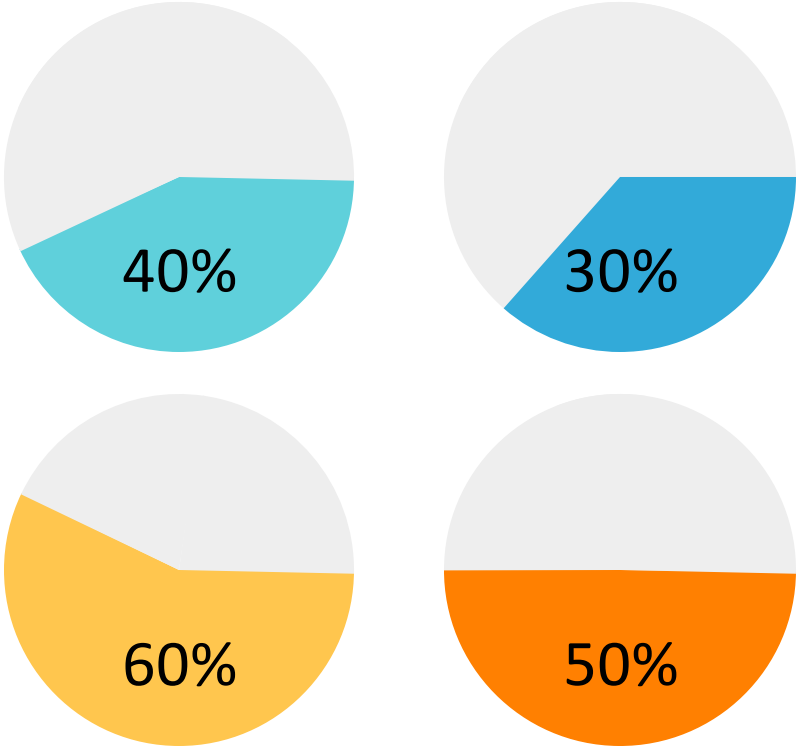
Mercury is a small planet

## Variability

Neptune is far away from Earth



# Big Data Infographics




### Volume

Venus has a beautiful name and is the second one from the Sun

### Variety

Mercury is the closest planet to the Sun and the smallest one

### Veracity

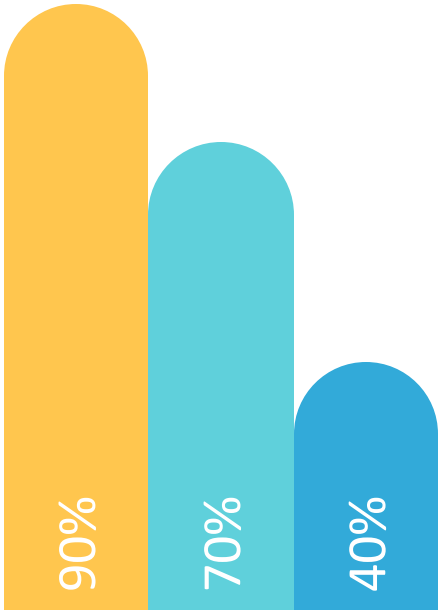
Despite being red, Mars is a cold place full of iron oxide dust

### Value

Jupiter is a gas giant and the biggest planet of them all

# Big Data Infographics

Insight 1




### Mercury

Planet Mercury is the smallest one in the entire Solar System

### Mars

Despite being red, Mars is actually a cold place full of iron oxide dust

### Jupiter

Planet Jupiter is a gas giant and the biggest one in the Solar System

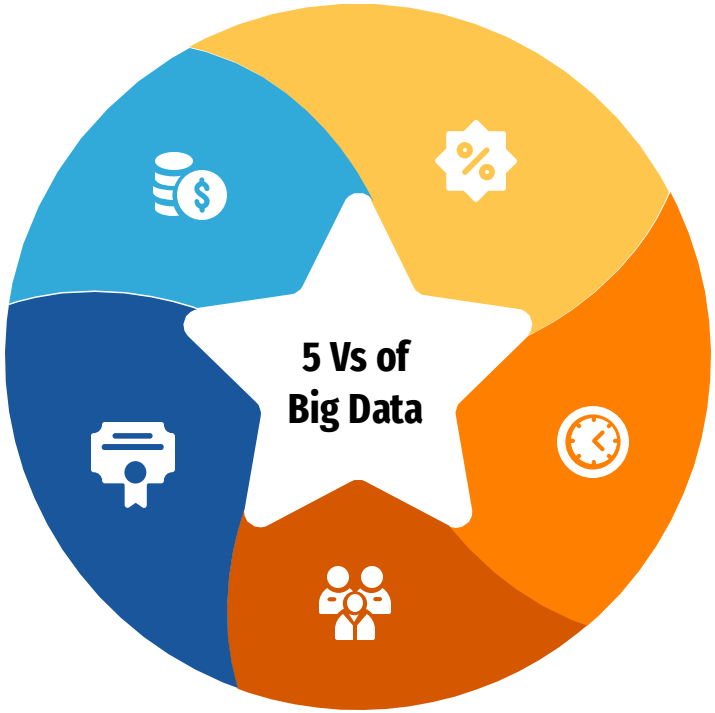
90% Mercury

70% Mars

40% Jupiter

Insight 2

# Big Data Infographics




**01**
**Volume**

Jupiter is the biggest planet of them all

**02**
**Velocity**

Venus is the second planet from the Sun

**03**
**Variety**

Despite being red, Mars is a cold place

**04**
**Veracity**

Earth is the planet where we live on

**05**
**Value**

Mercury is the closest planet to the Sun

# Big Data Infographics

## Mars

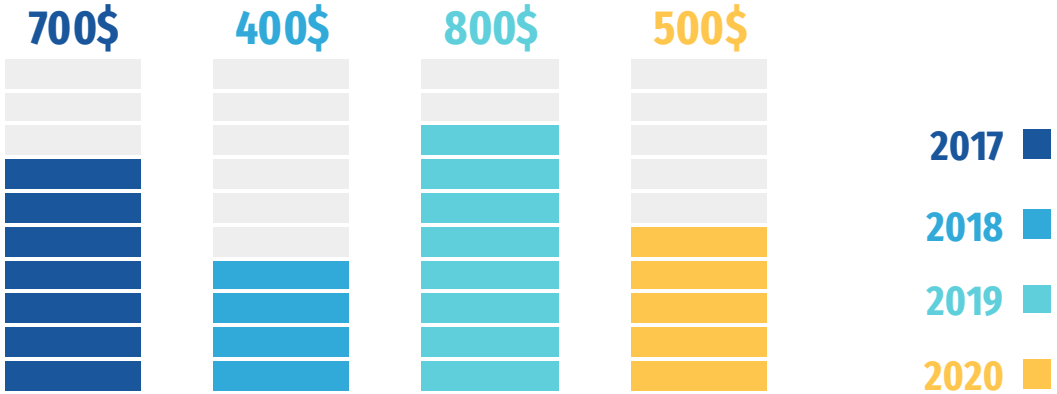
Despite its reddish cast, it’s actually an incredibly cold place

## Mercury

Mercury is the closest planet to the Sun and the smallest one of them all

## Saturn

Saturn is a planet composed mostly of hydrogen and helium, and has rings



# Big Data Infographics

**Low Volume**

Mercury is the closest planet to the Sun

**High Volume**

Jupiter is the biggest of them all

**Batch**

Despite being red, Mars is a cold place

Small Data vs. Big Data

**Real-Time**

Venus is the second planet from the Sun

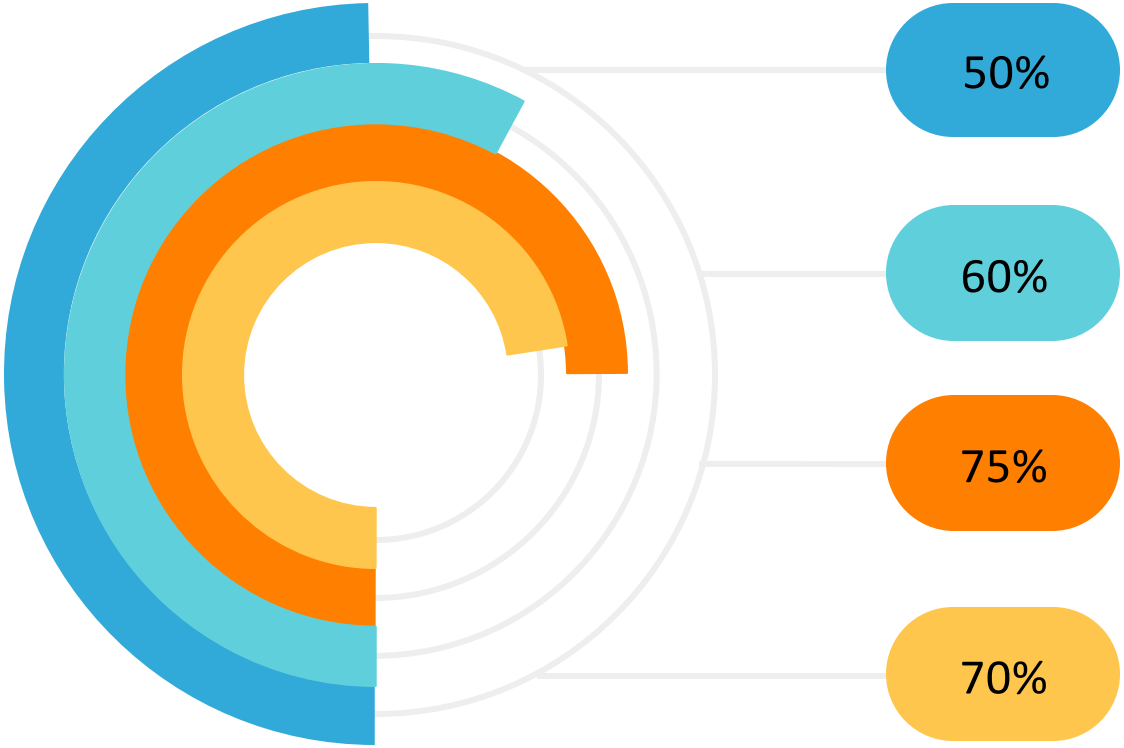
**Structured**

Earth is the planet where we live on

**Multi-Structured**

Saturn is the ringed one. It's a gas giant

# Big Data Infographics




**Volume**

Venus has a beautiful name and is the second one from the Sun

**Variety**

Mercury is the closest planet to the Sun and the smallest one

**Veracity**

Despite being red, Mars is a cold place full of iron oxide dust

**Value**

Jupiter is a gas giant and the biggest planet of them all

# Big Data Infographics

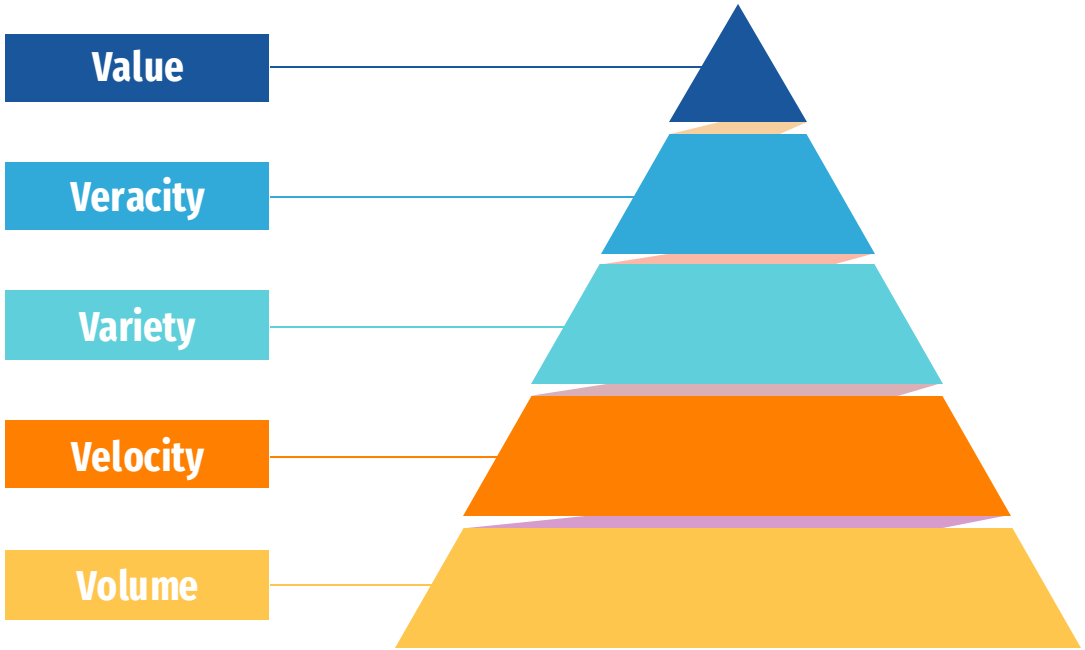
Despite being red, Mars is actually a cold place

Mercury is the smallest planet in the Solar System

Earth is the third planet from the Sun and the one with life

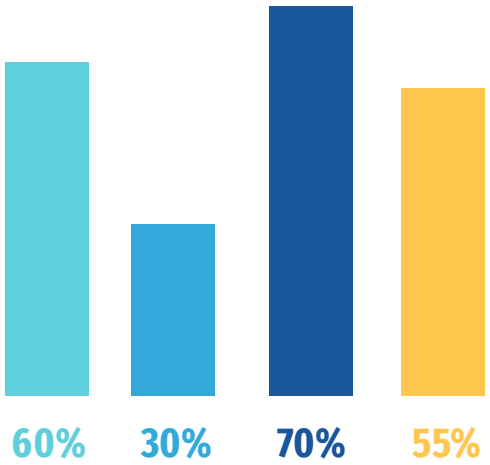
Neptune is the farthest planet from the Sun and a gas giant

Jupiter is a gas giant and the biggest planet of them all

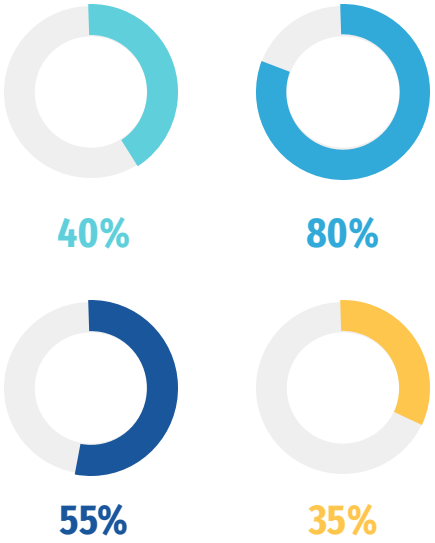


# Big Data Infographics

**Data 1**




**Data 2**



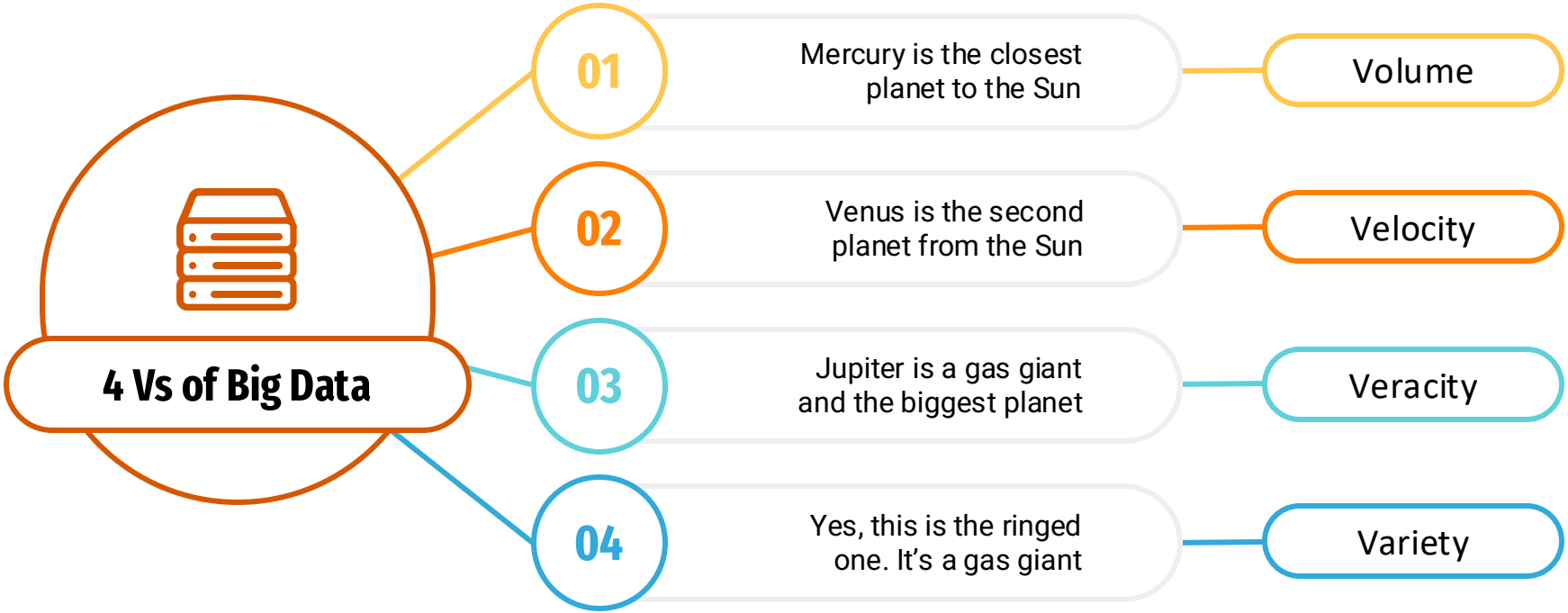

**01** Mercury is the smallest planet in the Solar System

**02** Mars is actually a cold place, it's full of iron oxide dust

**03** Jupiter is a gas giant and the biggest of them all

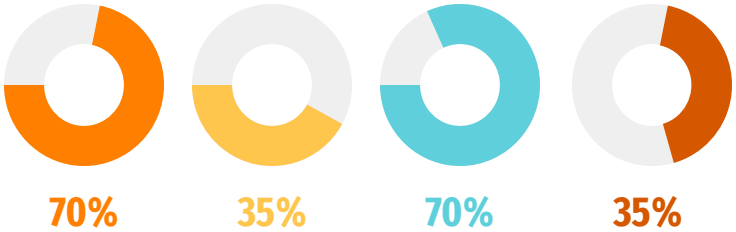
**04** Saturn is composed of hydrogen and helium

# Big Data Infographics

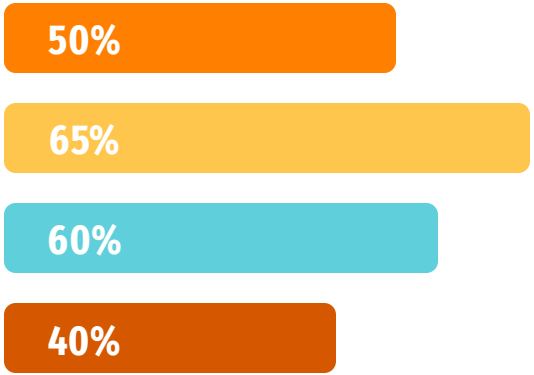


# Big Data Infographics

Data 1




Data 2




### Venus

Venus has a beautiful name and is the second planet from the Sun

### Mercury

Mercury is the closest to the Sun and the smallest one

### Mars

Despite being red, it's a cold place. It's full of iron oxide dust

### Jupiter

Jupiter is a gas giant and the biggest planet in the Solar System

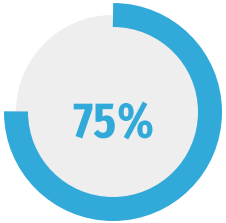
# Big Data Infographics

## Mercury

Mercury is the closest planet to the Sun and the smallest one in the Solar System

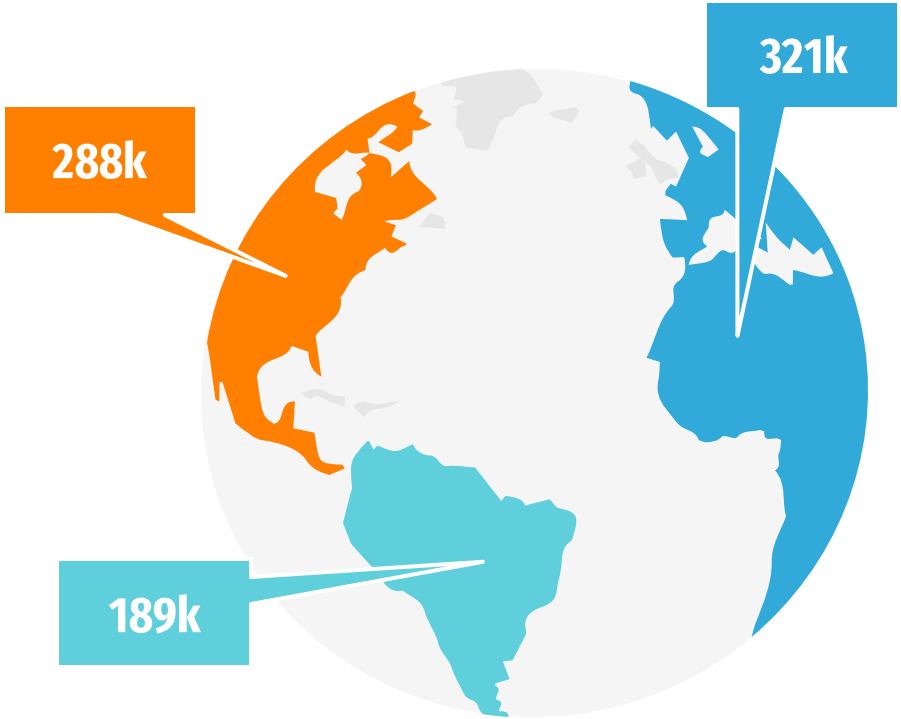
## Neptune

Neptune is the farthest planet from the Sun. It's the fourth-largest by diameter




## Mars

Despite being red, Mars is a cold place. The planet is full of iron oxide dust



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- **Change the color** by clicking on the paint bucket.
- Then **resize** the element by clicking and dragging one of the square-shaped points of its bounding box (the cursor should look like a double-headed arrow). Remember to hold Shift while dragging to keep the proportions.
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